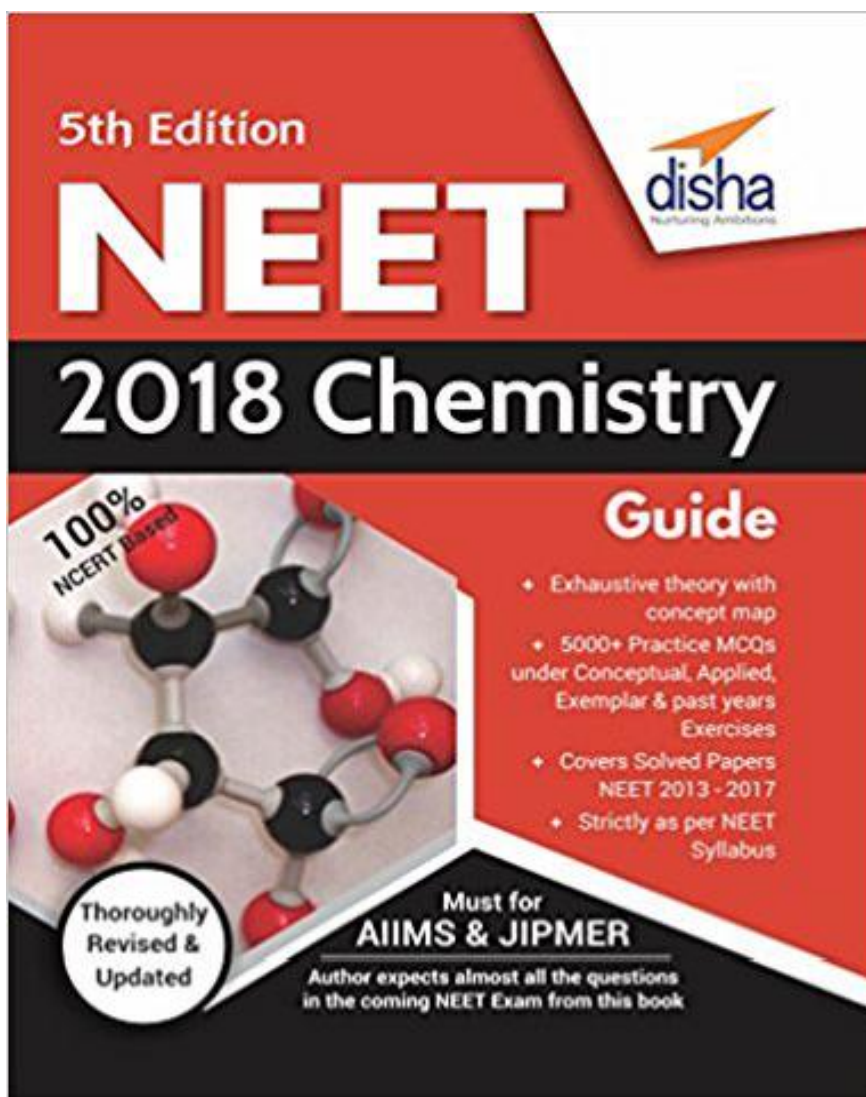


Organic Chemistry : Some Basic Principles and Techniques

This Chapter "Organic Chemistry : Some Basic Principles and Techniques" is taken from our Book:



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Chapter 12 Organic Chemistry : Some Basic Principles and Techniques

TETRAVALENCY AND SHAPES OF ORGANIC MOLECULES

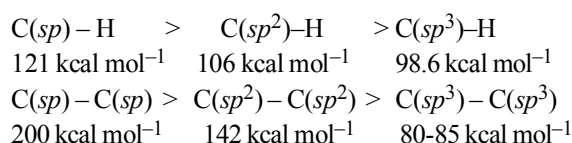
The tetravalence, formation of covalent bond and catenation property of carbon results in the formation of a large number of compounds of carbon. These compounds are covalent in nature and are studied under a branch of chemistry called organic chemistry. Carbon atoms present in organic compounds are either

sp , sp^2 or sp^3 hybridised.

Hybridisation influences the bond length and bond strength in organic compounds. The sp hybrid orbital contains more s-character and hence, it is closer to its nucleus and forms shorter and stronger bonds than sp^3 hybrid orbital.

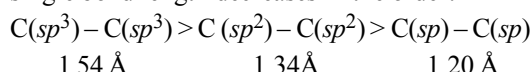
The bond angles and geometry associated with the three types of hybridization are summarized below :

Hybridisation	sp^3	sp^2	sp
Angle	$109^\circ 28'$	120°	180°
Geometry	Tetrahedral	Trigonal	Linear
Example	Alkane, Cycloalkane and in saturated part of all organic molecules	Alkenes and other compounds containing C=C, C=O, C=N and C=S double bonds	Alkynes and all other compounds containing C≡C and C≡N triple bonds,
Bond	Four- σ	Three- σ One- π	Two- σ Two- π
s%	25	33.3	50
p%	75	66.7	50
Electronegativity	2.48	2.75	3.25



Note : (i) Greater is the s - character of the hybrid orbitals, greater is the electronegativity. Thus, a C - atom having an sp hybrid orbital with 50% s - character is more electronegative than sp^2 (33% s-character) or sp^3 (25% s-character) hybrid orbitals.

(ii) Since a bigger orbital forms a longer bond, therefore, C-C single bond length decreases in the order:



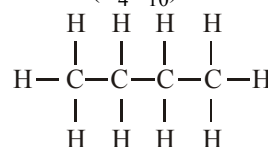
STRUCTURAL REPRESENTATIONS OF ORGANIC COMPOUNDS

(i) **Complete, Condensed and Bond - line Structural Formulas :**

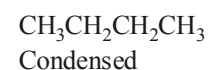
For example few compounds are represented in all three formulas below :

Ex:

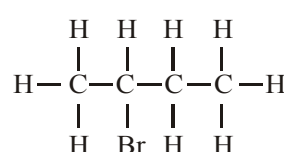
(a) Butane (C_4H_{10})



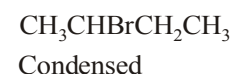
Complete



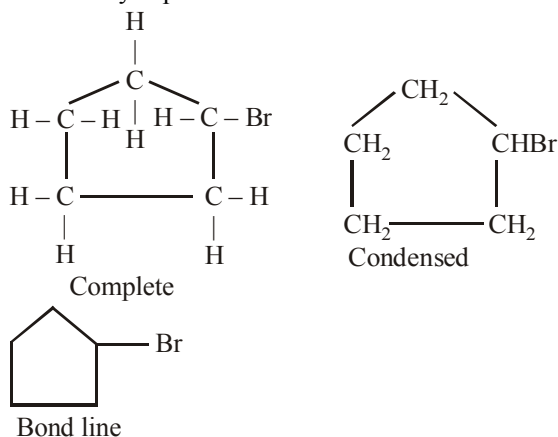
(b) 2 - Bromobutane



Complete



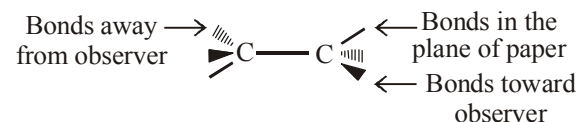
(c) 1-Bromocyclopentane



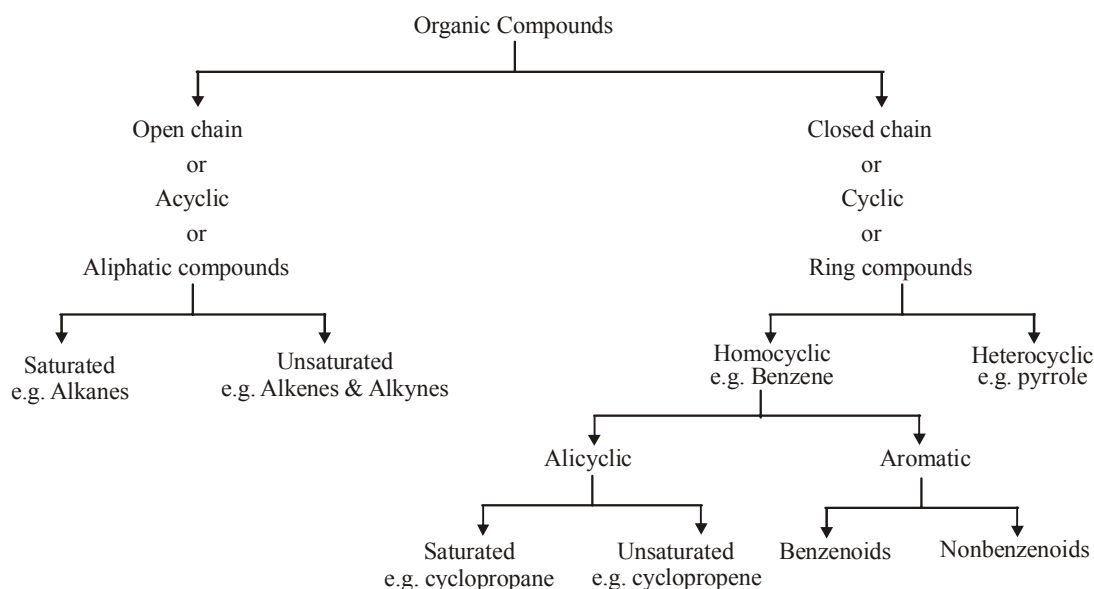
(ii) 3D Representation

By using solid ($\blacktriangle$) and dashed ($\dashv$) wedge formula, 3-D image of a molecule can be shown from 2D picture. The dashed line depicts the bond projecting out of the plane of paper and away from the observer while solid-wedge depicts the bond projecting out of the plane of paper and towards the observer. The bonds lying in plane of paper are depicted by using a normal line (—).

Ex: Ethane



CLASSIFICATION OF ORGANIC COMPOUNDS



NOMENCLATURE OF ORGANIC COMPOUNDS

Nomenclature of Alkanes

(i) Straight chain hydrocarbons:

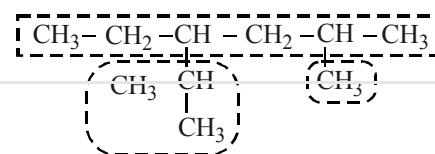
The names of such compounds are based on their chain structure, and end with suffix 'ane' and carry a prefix indicating the number of C - atoms present in the chain. The alkanes differ from each other by no. of $-\text{CH}_2$ groups in the chain.

Some names are:

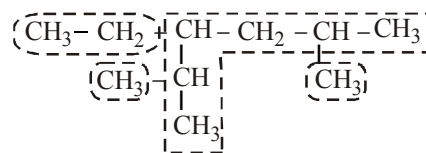
Methane	CH_4	Butane	C_4H_{10}
Ethane	C_2H_6	Pentane	C_5H_{12}
Propane	C_3H_8	Hexane	C_6H_{14}

(ii) Branched chain hydrocarbons: The rules for naming are:

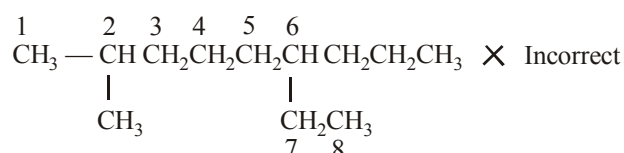
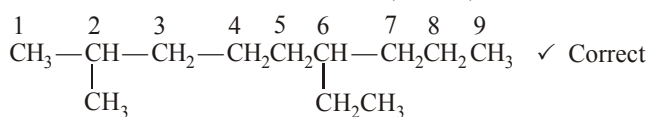
- (a) The longest C - chain in the molecule is identified. Out of all the possible chains. If there are the possibilities of more than one longest chain the chain which possesses more than one side chain is selected.



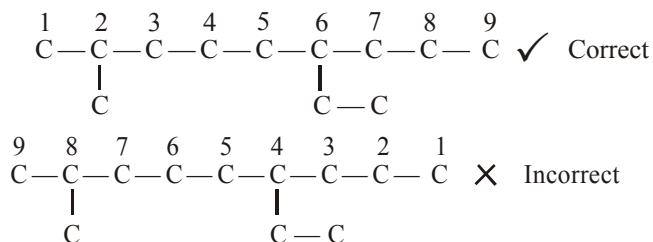
2 side chains (wrong)



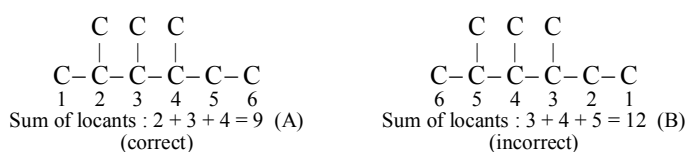
3-side chains (correct)



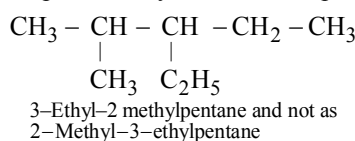
- (b) The numbering is done in such a way that branched carbon atoms get the lowest possible numbers.



In case, the parent chain has two or more substituents, numbering must be done in such a way that the sum of the locants on the parent chain is the lowest possible. Thus in following structures A and B numbering may be done in two ways : in one (A), the sum of locants is 9 while in other (B), it is 12, hence the former is correct while the latter is wrong.

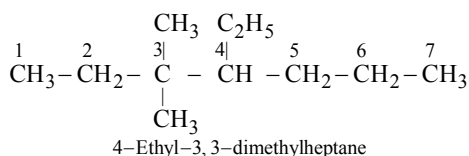
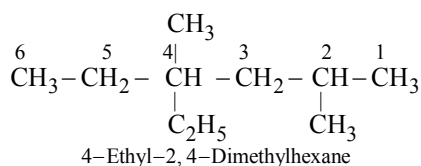


- (c) The names of alkyl groups attached as a branch are prefixed to the name of parent alkane and position of substituents is indicated by appropriate numbers. If different alkyl groups are present they are listed in alphabetical order. For example



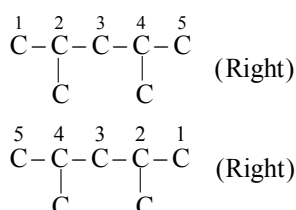
- (d) If two or more identical substituent groups are present, then numbers are separated by commas. The names of identical substituents are indicated with prefixes such as di, tri, tetra, penta, hexa, etc. While writing names of substituents spelling of these prefixes are not considered.

For example :



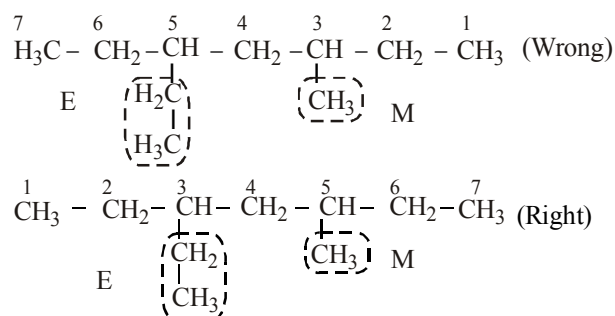
- (e) If side chains are at same carbon from both sides, following rules are adopted :

- (i) For same side chains numbering may be given from any end.

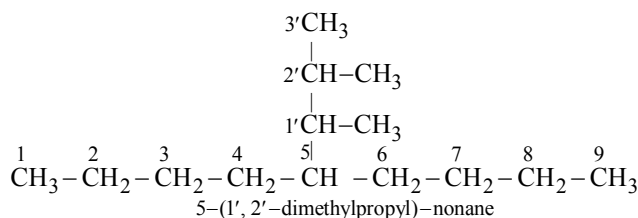


- (2) If side chains are different numbering is given in alphabetical preference order.

Ex.



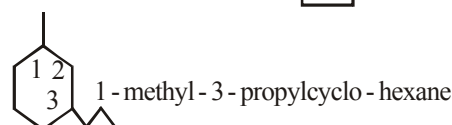
(f) Naming the complex substituent : In case the substituent on the parent chain is complex (i.e., it has itself branched chain) then it is named as a substituted alkyl group and its carbon chain is numbered from the carbon atom attached to the main chain. The name of this complex substituent is written in bracket to avoid confusion with the numbers of the main chain. e.g.,



Nomenclature of Cyclic Compounds

A saturated monocyclic compound is named by prefixing 'cyclo' to the corresponding straight chain alkane. If side chains are given, then rules given above are applied.

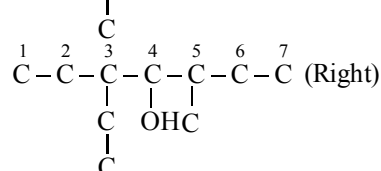
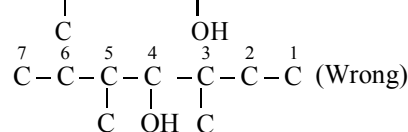
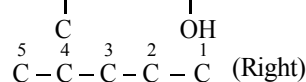
For example : Cyclobutane 



Nomenclature of Organic Compounds Containing Functional Groups

- (i) Numbering is given from the end from which functional group is closest.
- (ii) If the functional group is situated at same carbon from both sides, follow the rules of saturated hydrocarbons.

Ex. $\text{C}-\text{C}-\text{C}-\text{C}-\text{C}$ (Wrong)

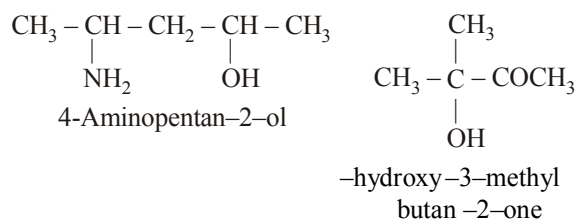


- (iii) If the organic compound contains two or more functional groups, one of the group is selected as principal functional group containing longest possible chain of carbon atom and all the remaining functional groups are treated as substituents.

Order of preference used while selecting the principal functional group

Carboxylic acids > Sulphonic acids > anhydrides > esters > acid chlorides > acid amides > nitriles > aldehydes > cyanides > isocyanides > ketones > alcohols > phenols > thiols > amines > ethers > alkenes > alkynes.

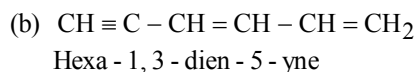
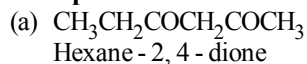
For example :



- (iv) — R, — C₆H₅, halogens (F, Cl, Br, I), —NO₂, —OR etc. are always prefix substituents.

- (v) If more than one functional group of same type is present, their number is indicated by adding di-, tri, etc. before the class suffix.

Examples :



See the following table for IUPAC names of certain organic compounds

Class of compounds	Functional group structure	Functional group prefix	Functional group suffix
Alkanes	—	—	-ane
Alkenes	>C=C<	—	-ene
Alkynes	—C≡C—	—	-yne
Arenes	—	—	-
Halides	—X (X = F, Cl, Br, I)	halo	-
Alcohols	—OH	hydroxy	-ol
Aldehydes	—CHO	formyl, or oxo	-al
Ketones	>C=O	oxo	-one
Nitriles	—CN≡	cyano	nitrile
Ethers	R-O-R	alkoxy	-
Carboxylic acids	—COOH	carboxy	-oic acid
Carboxylate ions	—COO [—]	—	-oate
Esters	—COOR	alkoxycarbonyl	-oate
Acyl halides	—COX (X = F, Cl, Br, I)	halocarbonyl	-oyl halide
Amines	—NH ₂ , >NH, >N—	amino	-amine
Amides	—CONH ₂ —CONHR, —CONR ₂	carbamoyl	-amide
Nitro compounds	—NO ₂	nitro	-
Sulphonic acids	—SO ₃ H	sulpho	sulphonic acid
Thio alcohol	—SH	Mercapto—	—thiol
Anhydride	$\begin{array}{c} \text{—CO—} \\ \text{—CO—} \end{array} \text{O}$	—	—oic anhydride
Isocyanide	—N≡C	Carbylamine	—iso nitrile

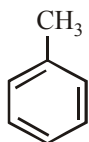
Note : $\text{--}\overset{\text{O}}{\parallel}\text{C--H}$, $\text{--}\overset{\text{O}}{\parallel}\text{C--OH}$, $\text{--}\overset{\text{O}}{\parallel}\text{C--Cl}$, $\text{--}\overset{\text{O}}{\parallel}\text{C--NH}_2$ & $\text{--C}\equiv\text{N}$

are the groups of which carbons are also included in the longest chain. If these are alone in a compound they come on first carbon, if double, comes on first and last positions.

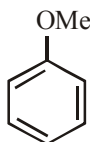
Nomenclature of Aromatic Compounds

- (i) The substituent is placed as prefix to the word benzene.

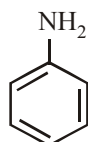
Examples :



Methyl benzene
(Toluene)



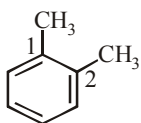
Methoxy benzene
(Anisole)



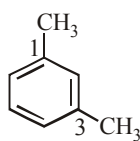
Amino benzene
(Aniline)

- (ii) If benzene ring is disubstituted, position of substituents is defined by numbering the C-atoms of ring such that substituents are located at lowest numbers possible.

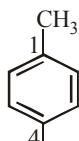
Example :



1,2-Dimethylbenzene
(*o*-Xylene)



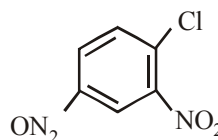
1,3-Dimethylbenzene
(*m*-xylene)



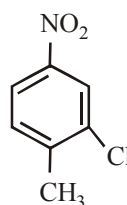
1,4-Dimethylbenzene
(*p*-xylene)

- (iii) For tri - or higher substituted benzene derivatives, the compounds are named by identifying substituent positions on the ring by following lowest locant rule. In some cases, common name of benzene derivatives is taken as base compound. Substituent of base compound is assigned no. 1 and then the direction of numbering is chosen such that the next substituent gets the lowest number. The substituents should be numbered according to alphabetical order.

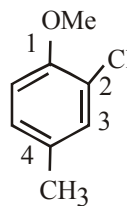
ex:



1 - Chloro - 2, 4 - dinitrobenzene



2 - Chloro - 1 - methyl - 4 - nitrobenzene

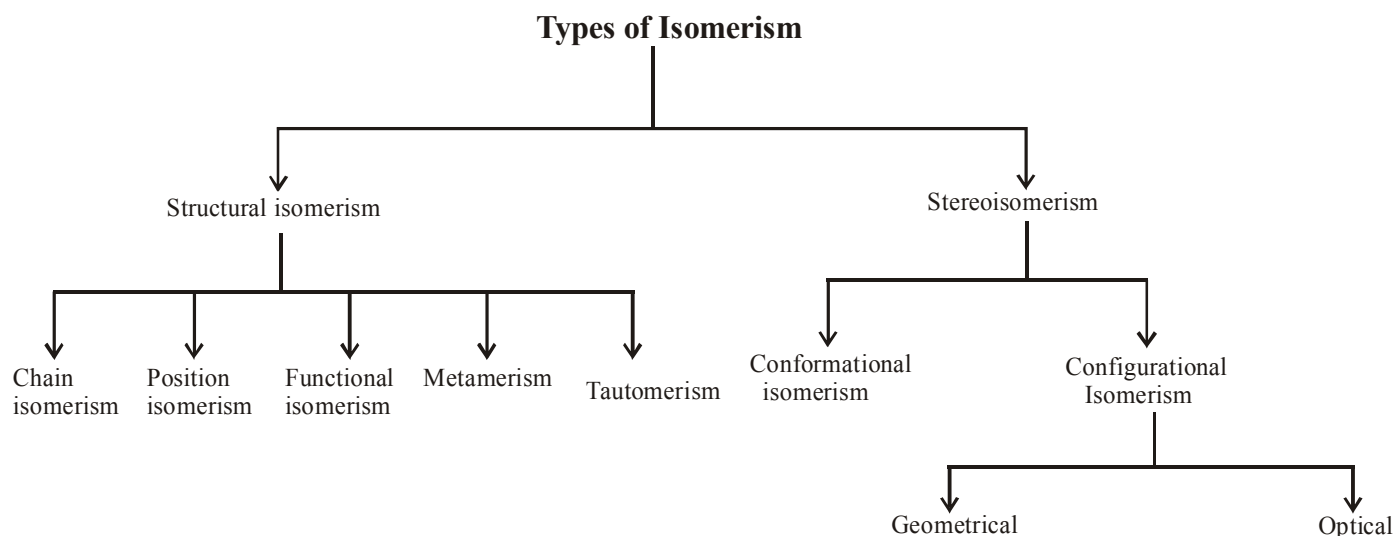


2 - Chloro - 4 - methyl anisole

- (iv) When benzene ring is attached to an alkane with a functional group, it is considered as substituent instead of parent and its name as a substituent is phenyl

ISOMERISM

The phenomenon of existence of two or more compounds possessing same molecular formula but different properties is called isomerism and the compounds are called isomers. The following chart shows different types of isomerism.



Structural Isomerism: It arises when compounds have same molecular formula but different structures. It arises because of the difference in the sequence of covalently bonded atoms in the molecule without reference to space.

Types of Structural Isomerism

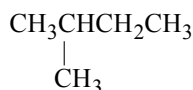
- (i) **Chain isomerism** : Two or more compounds with same molecular formula but different carbon skeletons are called chain isomers and the phenomena is known as chain isomerism.

Examples :

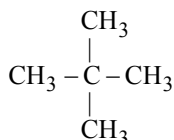
Pentane : C_5H_{12}



n-Pentane



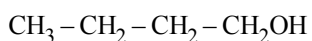
Isopentane



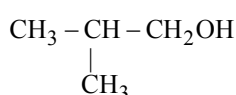
Neopentane

n-Pentane, isopentane and neopentane possess the chain of five, four and three carbons, respectively. Hence they are chain isomers.

Butyl alcohol : C_4H_9OH



n-Butyl alcohol



Isobutyl alcohol

These two butyl alcohols are chain isomers.

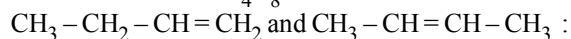
- (ii) **Position isomerism** : It is shown by the compounds in which there is difference in the position of attachment of functional group, multiple bond or substituent along the same chain length of carbon atoms.

Examples : $CH_3 - CH_2 - CH_2Cl$ & $CH_3 - \underset{\substack{| \\ Cl}}{CH} - CH_3$

1-Chloropropane

2-Chloropropane

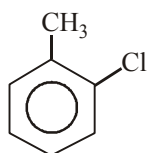
Molecular formula : C_4H_8



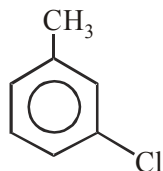
1-Butene

2-Butene

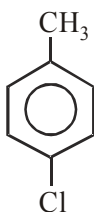
Note : In the disubstituted benzene derivatives also position isomerism exists because of the relative position occupied by the substituents on the benzene ring. Thus, Chlorotoluene, $C_6H_4(CH_3)Cl$ exists in three isomeric forms - ortho, meta and para.



o-Chlorotoluene



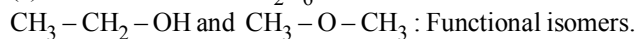
m-Chlorotoluene



p-Chlorotoluene

- (iii) **Functional group isomerism** : When two or more compounds have same molecular formula but different functional groups. Such compounds are called functional group isomers

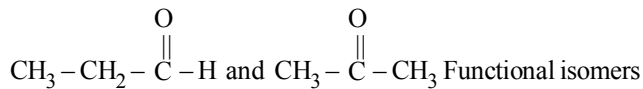
(a) Molecular formula : C_2H_6O



Ethyl alcohol Dimethyl ether

(Alcohol) (Ether)

(b) Molecular formula : C_3H_6O



Propanal
(Aldehyde)

Propanone
(Ketone)

(c) Molecular formula : $C_3H_6O_2$



:Functional isomers

Propanoic acid
(Acid)

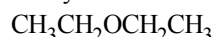
Methyl acetate
(Ester)

- (iv) **Metamerism** : This type of isomerism is due to unequal distribution of substituents on either side of the polyvalent functional group.

Members belong to the same homologous series.

Ex.

(a) Diethyl ether and methyl propyl ether



Diethyl ether

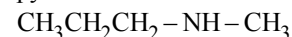


Methyl propyl ether

(b) Diethyl amine and methyl propyl amine



Diethyl amine



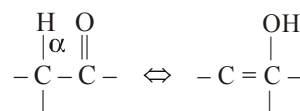
Methyl propyl amine

- (v) **Tautomerism** : Tautomerism is a special type of functional group isomerism which arises due to the migration of H-atom as proton from a polyvalent atom to other polyvalent atom with reshuffling of π bond.

Keto-Enol Tautomerism :

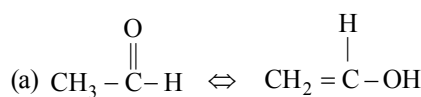
When the tautomers exist in the two forms keto & enol then, such type of tautomerism is called keto-enol tautomerism.

Keto means the compound has a keto group $>C=O$, and the enol form has both double bond and OH (hydroxy) group joined to the same carbon.



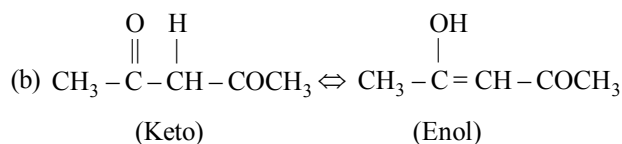
Keto form

Enol form



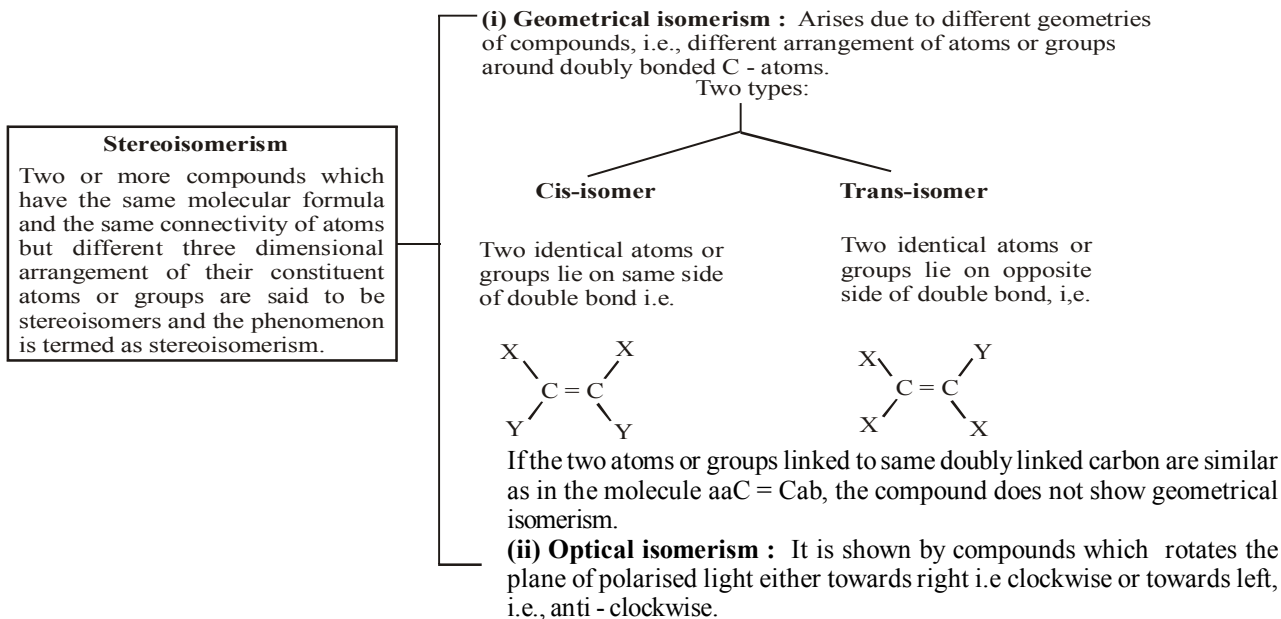
Acetaldehyde

Vinyl alcohol



(Keto)

(Enol)



Optical activity : It is the ability of a substance to rotate the plane of polarisation of plane polarised light.

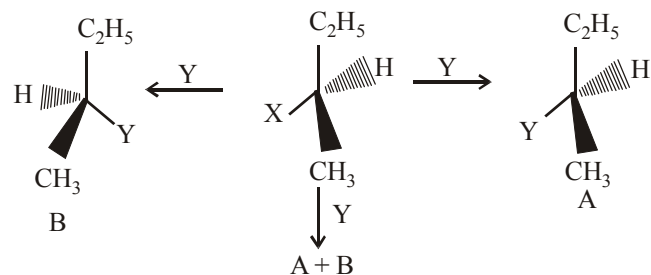
- One rotating the plane of polarised light to the left. This form is named as laevorotatory. (Latin, laevous = left) or direction (–) form
- One rotating the plane of polarized light exactly to the same extent but to the right is named dextrorotatory (Latin Dexter – right) or direction (+) – form.
- An inactive form which does not rotate the plane of polarized light at all. This is a mixture of equal amounts of (+) and (–) – forms and hence it is optically inactive. It is named (±) - mixture or Racemic mixture. (Latin, Racemic - mixture of equal compounds) **Asymmetric carbon atom :** A carbon atom attached to four different atoms and groups is called asymmetric carbon atom. e.g. $CH_3^*CHOHCOOH$.

Chirality : If the mirror image of the molecule is different from the molecule it is said to be a chiral molecule. In such case if one configuration of the molecule is placed above its mirror image configuration, the similar atoms and groups do not fall over each other and the configurations are said to be *nonsuperimposable*. Chirality is the necessary condition for a compound to be optically active.

Enantiomers : Pairs of nonsuperimposable mirror images are called enantiomers. Enantiomers are non superimposable mirror image isomers. They have identical physical properties (bpt, mpt, density etc.) except for their rotation of plane polarised light. They are much more difficult to separate. In nature very often only one enantiomer is produced. Living organisms are one of the best sources of optically active compounds (plants, enzymes, animals, cells etc.).

Diastereo Isomers : Stereo isomers which are not mirror images of each other are called diastereo isomers. They have different physical properties (mpt, bpt, solubility) and are often easy to separate by distillation, recrystallisation, chromatography etc.).

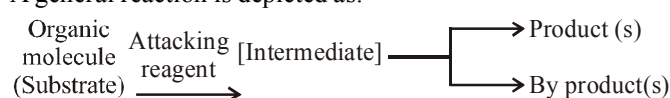
Racemimixture : A mixture containing equal amounts of two enantiomers will have zero optical rotation and is called *racemic mixture*



if only (A) is obtained, process is called retention of configuration. If only (B) is obtained, process is called inversion of configuration. If 50:50 mixture of above two is obtained then the process is called racemisation and the product is optically inactive.

REACTION MECHANISM

A general reaction is depicted as:

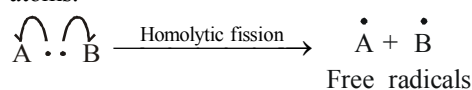


A sequential account of each step, describing details of electron movement, energetics during bond cleavage and bond formation, and the rates of transformation of reactants into products is referred to as reaction mechanism.

Fission of a Covalent Bond

A covalent bond can be cleaved in two ways:

- Homolytic cleavage :** On bond breaking, one of the electrons of shared pair in covalent bond goes with each of the bonded atoms.

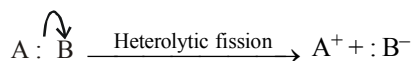


It results in the formation of free radicals which contain an odd or unpaired electron. Such reactions are called free radical or homopolar or non-polar reactions.

Favourable conditions :

- High temperature
- Light of suitable wavelength
- Non polar solvent
- Presence of peroxide or oxygen
- Ability of substrate and attacking reagent to produce free-radicals.

- (ii) **Heterolytic cleavage** : When a covalent bond joining two atoms A and B breaks in such a way that both the electrons of the covalent bond (i.e., shared pair) are taken away by one of the bonded atoms, the mode of bond cleavage is called heterolytic fission or heterolysis.



(when B is more electronegative than A)

It results in formation of cations and anions

Favourable conditions : (a) Low temperature (b) Polar solvent (c) Presence of acid or base catalyst (d) Polar nature of the substrate and attacking reagent.

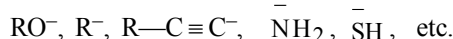
Nucleophiles and Electrophiles

A reagent that brings an electron pair is called a nucleophile i.e. nucleus seeking and the reaction is called nucleophilic.

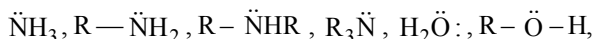
They are of two types :

(a) Charged Nucleophiles :

All the negative ions qualify as nucleophiles.



(b) Neutral Nucleophiles : Central atom has electron pair.

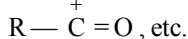
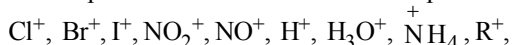


A reagent that takes away an electron pair is called an electrophile (E^+) i.e., electron seeking and the reaction is called electrophilic. Electrophiles may be neutral or positively charged.

(a) Neutral electrophiles : Central atom has deficiency of electrons. ex. $BF_3, BCl_3, AlCl_3, BeCl_2, FeCl_3, SO_3$, etc..

(b) Charged electrophiles :

All the positive ions behave like electrophiles.



Electrophiles are generated by heterolysis of a covalent bond.

Transition metal cations are electrophiles

For example : $Fe^{3+}, Fe^{2+}, Ag^+, Hg^{2+}, Cd^{2+}$, etc.

Electron Displacement

The electron displacement in an organic molecule may take place either in ground state under the influence of an atom or a group or in presence of an appropriate attacking reagent.

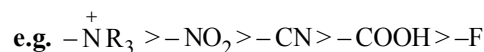
Electron displacement due to influence of an atom or group present in molecule cause permanent polarisation of the bond. Example: Inductive effect and resonance effect.

Temporary electron displacement effects are seen in a molecule when a reagent approaches attack it.

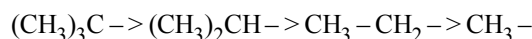
Example: electromeric or polarisability effect.

- (i) **Inductive Effect** : The polarisation of σ - bond caused by polarisation of adjacent σ - bond is called inductive effect. It is a permanent effect. The effect decreases as the number of intervening C - atoms. Inductive effect (I-effect) is of two types.

- (a) **-I effect** : Atoms or group of atoms which attract the bonded electrons more strongly than hydrogen atom, are said to have -I effect and are termed as electron attracting (-I groups). Such groups when linked with a carbon chain make it electron poor.



- (b) **+I Effect** : Those atoms or group of atoms which attract the shared electron pair (bond pair) less strongly than hydrogen atom are said to have +I (electron repelling) effect. Such groups when attached with a carbon chain displace the shared σ electrons towards the chain and make it electron rich.



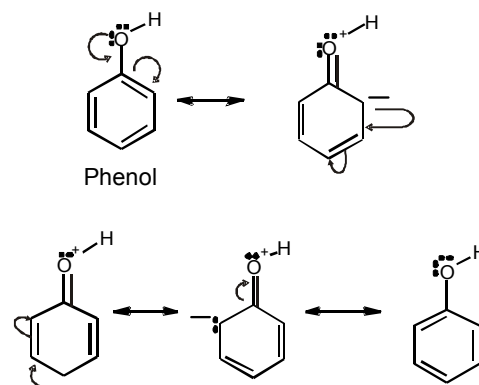
- (ii) **Resonance effect** : It is defined as the polarity produced in a molecule by the interaction of two π - bonds or between a π - bond and lone pair of e^- s present on an adjacent atom. It is also called mesomeric effect. It is a permanent effect.

Two types of resonance effects:

(a) Positive resonance effect (+ R effect)

In this effect, transfer of electrons is away from an atom group attached to the conjugated system. This effect results into increase in electron density at certain positions in molecule.

Example : Phenol



Examples : halogen, $-OH, -OR, -OCOR, -NH_2, -NHR, -NR_2, -NHR, -NR_2, -NHCOR$

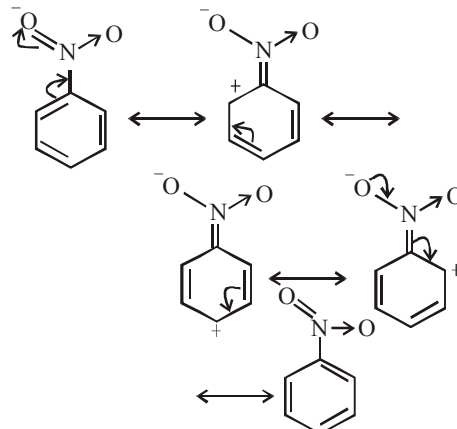
Decreasing order of activating (+R) groups

$-O-, -NH_2, -NHR, -NR_2, -OH, -OR, -NHCOR, -R, -Ar, -X$ (Till $-OH$ they are very strong activating groups)

(b) Negative resonance effect (- R effect)

In this effect, transfer of electrons occurs towards atom or group attached to the conjugated system.

Example : Nitrobenzene



Examples : $-\text{COOH}$, $-\text{CHO}$, $\text{C}=\text{O}$, $-\text{CN}$, $-\text{NO}_2$

The very strong deactivating ($-\text{R}$) groups are

$-\text{NO}_2$, $-\text{CF}_3$, $-\text{NR}_3^+$

Conditions for Resonance :

- (1) The atomic arrangement is the same in all the canonical forms.
- (2) Same number of paired or unpaired electrons must be present in each canonical form.
- (3) The canonical forms must possess same or nearly same energy.
- (4) The molecule must have a planar structure.

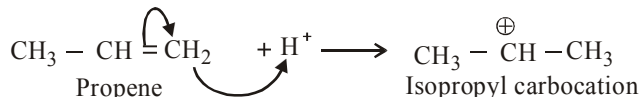
(iii) **Electromeric effect (E - Effect)** : It is defined as the complete transfer of a shared pair of π - electrons to one of the atoms (more electronegative) joined by a multiple bond on demand of an attacking reagent. It is a temporary effect.

Two types of electromeric effect:

(a) +E effect

In this, π - electrons of multiple bond are transferred to that atom to which reagent gets attached.

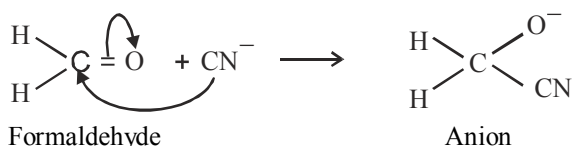
For example :



(b) -E effect

In this, π - electrons of multiple bond are transferred to that atom to which attacking reagent does not get attached

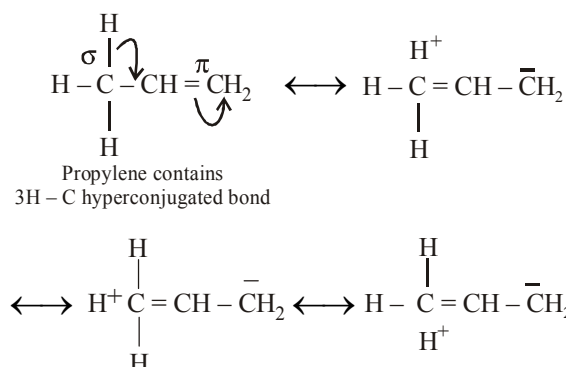
For example :



Oxygen being more electronegative pulls π electrons of multiple bond toward itself. Carbonyl C acquires positive charge thus got attacked by nucleophile.

(iv) **Hyperconjugation** : It involves delocalisation of σ - electrons of C - H bond of an alkyl group directly attached to an atom of unsaturated system or to an atom with an unshared p - orbital. It is a permanent effect. Hyperconjugation is also called no - bond resonance.

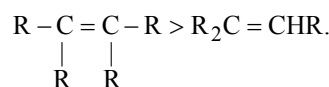
In propylene, there are three H - C sigma bonds in conjugation with the π bond. So, only three resonating forms can be written.



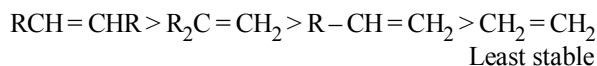
Effects of hyperconjugation :

(1) Stability of alkenes :

(a) Alkylated alkenes are more stable than others.



[Tetraalkyl ethylene]



(b) Greater the number of α C - H bonds in an alkene, greater will be the number of hyperconjugated structures, hence greater will be the stability.

(2) **Bond lengths** : Hyperconjugation in a molecule results in a change in bond lengths.

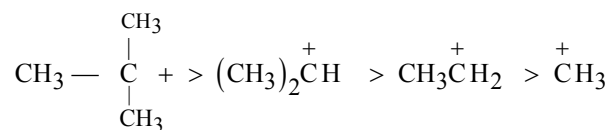
(3) Stability of alkyl free radicals :

t -alkyl > $sec.$ alkyl > primary alkyl > CH_3 (methyl radical)

Greater the number of hyperconjugative structures of an alkyl radical, higher is the stability.

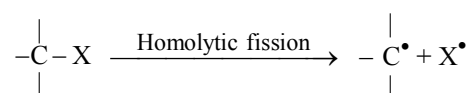
(4) **Stability of alkyl carbonium ions** : Greater the number of alkyl groups attached to positively charged carbon atom, greater is the hyperconjugation interaction and stabilisation of the cation.

The relative stability of carbocations follows the order:



INTERMEDIATES

Free Radicals



If $\text{EN of C} \approx \text{EN of X}$

It is an atom or group of atoms possessing an odd or unpaired electron. It is electrically neutral and shows paramagnetism.

Stability of free radicals is due to following reasons :

(i) Resonance (ii) Hyperconjugation

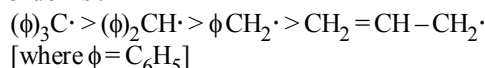
Order of stability of free radicals :

t -butyl > isopropyl > ethyl > methyl

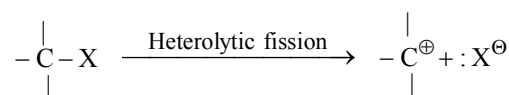
Stability on the basis of resonance :

Stability $\propto$ Number of resonating structures

The order is :



Carbonium ion or Carbocation $-\overset{\oplus}{\text{C}}-$

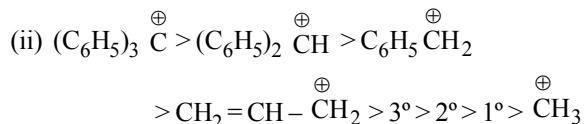
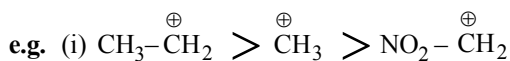


If $\text{EN of X} > \text{EN of C}$

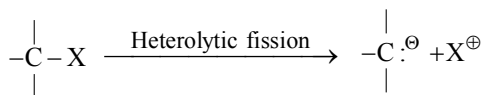
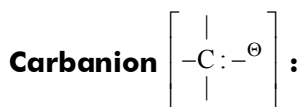
Organic species containing positively charged carbon atom is known as carbonium ion. The positively charged carbon atom contains six electrons in its valence shell.

Stability of carbonium ions :

$$\text{Stability of carbocation} \propto +I \propto \frac{1}{-I}$$



Stability $\propto$ No. of resonating structures



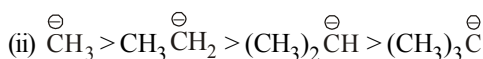
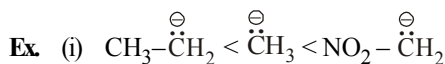
If EN of C > EN of X

An organic species containing negatively charged carbon atom is called carbanion.

Stability of carbanion is due to following reasons

- Inductive effect
- Resonance
- s-character in Hybridisation

$$\text{Stability of Carbanion} \propto -I, \propto \frac{1}{+I}$$



Note : Carbanions are nucleophiles.

TYPES OF ORGANIC REACTIONS

(i) Substitution or Displacement Reactions

Substitution or displacement reactions are those reactions in which an atom or group of atoms attached to a carbon atom in a substrate molecule is replaced by another atom or group of atoms.

(ii) Addition Reactions

Addition reactions are those in which the attacking reagent adds up to the substrate molecule. Such reactions are given by those compounds which possess double or triple bonds.

(iii) Elimination Reactions

The reverse of addition reactions are termed as elimination reactions. In these reactions generally atoms or groups from two adjacent carbon atoms in the substrate molecule are removed and multiple bond is formed.

(iv) Rearrangement Reactions

The reactions which involve the migration of an atom or group from one site to another within the molecule (nothing is added from outside and nothing is eliminated) resulting in a new molecular structure are known as rearrangement reactions.

PURIFICATION OF ORGANIC COMPOUNDS

A large number of methods are available for the purification of substances. The choice of method, however depends upon the nature of substance (whether solid or liquid) and the types of impurities present in it. Following are some of the important methods which are commonly employed for the purification of organic compounds:

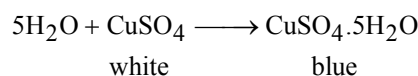
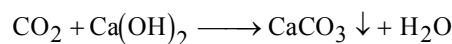
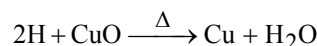
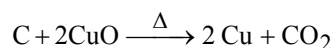
- (1) Filtration, (2) Crystallization, (3) Fractional crystallization, (4) Sublimation, (5) Distillation, (6) Fractional Distillation, (7) Distillation under reduced pressure, (8) Steam Distillation, (9) Differential extraction, (10) Chromatography

QUALITATIVE AND QUANTITATIVE ANALYSIS

Qualitative Analysis

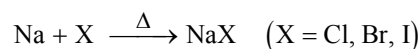
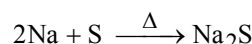
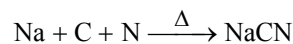
The elements present in organic compounds are C and H. In addition to these, they may also contain O, N, S, halogens and P.

Detection of C and H: They are detected by heating the compound with copper (II) oxide. C is oxidised to CO_2 and H to H_2O . CO_2 turns white lime water milky, while water formed turn white anhydrous CuSO_4 to blue hydrated CuSO_4 .



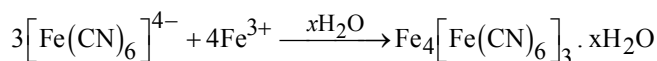
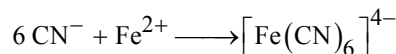
Detection of other elements

Preparation of Lassaigne's extract : N, S halogens and P are detected by Lassaigne's test. The elements present in organic compounds are converted to ionic form by fusing with Na-metal. The reactions involved are:

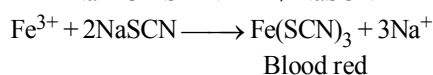


They are extracted from fused mass by boiling with distilled water. The extract is known as sodium fusion extract (Lassaigne's extract).

- (a) **Test for N :** The extract is boiled with FeSO_4 and then acidified with conc. H_2SO_4 . Prussian blue colour confirms presence of nitrogen.

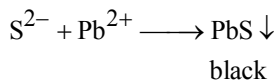


But if sulphur is also present along with nitrogen one gets blood red colouration due to formation of ferric thiocyanate.

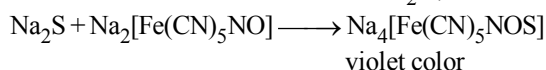
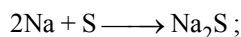


(b) Test for S:

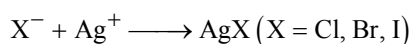
- (i) To extract, add CH_3COOH and lead acetate. A black ppt of PbS confirms S.



- (ii) Take small portion of Lassaigne's extract and add few drops of sodium nitroprusside solution, a violet colouration is obtained which fades away on standing.

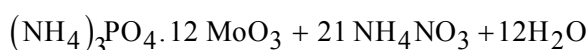
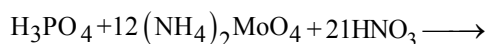
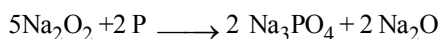
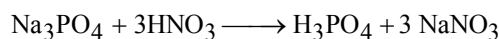


- (c) Test for halogens:** The extract is acidified with HNO_3 and then treated with AgNO_3 . A white ppt, soluble in NH_4OH indicates the presence of Cl. A yellowish ppt, sparingly soluble in NH_4OH indicates the presence of Br. A yellow ppt, insoluble in NH_4OH indicates the presence of I



If N or S is also present, then extract is first boiled with conc. HNO_3 to decompose CN^- and S^{2-} (they would otherwise interfere with AgNO_3 test for halogens).

- (d) Test for P:** Phosphorus is detected by fusing the organic compound with an oxidising agent like Na_2O_2 . P in the compound is oxidised to PO_4^{3-} which is then extracted with water. The solution is boiled with HNO_3 and then treated with ammonium molybdate. A yellow coloured ppt indicates presence of phosphorus.



Ammonium
phosphomolybdate

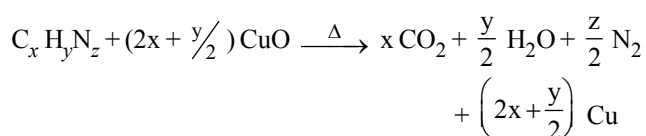
Quantitative Analysis

The percentage composition of elements present in an organic compound is determined in following ways:

(i) Carbon and Hydrogen

A known mass of an organic compound is burnt in presence of excess of O_2 and CuO , when C and H are oxidised to CO_2 and H_2O . Let mass of organic compound be 'W' g, mass of H_2O and CO_2 produced be W_1 and W_2 g. Then,

$$\text{Percentage of C} = \frac{12 \times W_2 \times 100}{44 \times W}$$



$$\text{Percentage of H} = \frac{2 \times W_1 \times 100}{18 \times W}$$

(ii) Nitrogen

- (a) Dumas method:** The N-containing organic compound, when heated with CuO in an atmosphere of CO_2 , yields free N_2 in addition to CO_2 and H_2O .

Let mass of organic compound = Wg

Volume of nitrogen collected = V_1 mL

Room temp = T_1 K

$$\text{Volume of N}_2 \text{ at STP} = \frac{P_1 V_1 \times 273}{760 \times T_1} \quad (\text{Let it be } V \text{ mL})$$

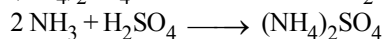
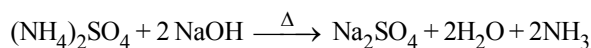
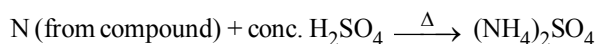
P_1 = Atm. pressure - Aqueous tension

22400 mL N_2 at STP weighs 28 g.

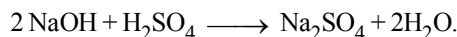
$$V \text{ mL N}_2 \text{ at STP weighs} = \frac{28 \times V}{22400} \text{ g}$$

$$\% \text{ of N}_2 = \frac{28 \times V \times 100}{22400 \times W}$$

- (b) Kjeldhal's method:** The compound is heated with conc H_2SO_4 when nitrogen gets converted to $(\text{NH}_4)_2\text{SO}_4$. The mixture is then heated with excess of NaOH . The liberated NH_3 gas is absorbed in an excess of standard solution of H_2SO_4 .



The volume of the acid left unused is found by titration against a standard alkali solution.



The difference between initial amount of acid taken and that left after reaction gives the amount of acid reacted with NH_3 .

$$\% \text{ of N} = \frac{14 \times M \times 2 \left(V - \frac{V_1}{2} \right)}{1000} \times \frac{100}{W}$$

Where M $\rightarrow$ molarity

V $\rightarrow$ volume of H_2SO_4 taken

V_1 $\rightarrow$ volume of NaOH used for titration of excess of H_2SO_4

W $\rightarrow$ mass of organic compound

$$2 \left(v - \frac{v_1}{2} \right) \rightarrow \text{volume of NH}_3 \text{ solution}$$

(iii) Halogens

Carius method: A known mass of organic compound is heated with fuming HNO_3 in presence of AgNO_3 . C and H in the compound are oxidised to CO_2 and H_2O . The halogen present forms AgX which is filtered, washed, dried and weighed.

Let mass of organic compound = Wg

mass of AgX formed = W_1 g

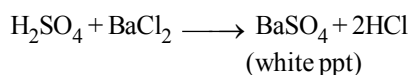
1 mol of AgX contains 1 mol of X

$$\text{mass of X in } W_1 \text{ g of Ag} = \frac{\text{atomic mass of X} \times W_1 \text{g}}{\text{molecular mass of AgX}}$$

$$\% \text{ of X} = \frac{\text{at. mass of X} \times W_1 \times 100}{\text{molecular mass of AgX} \times W}$$

(iv) **Sulphur**

A known mass of organic compound is heated with sodium peroxide or fuming HNO_3 . S in the compound is oxidised to H_2SO_4 . It is precipitated as BaSO_4 by adding excess BaCl_2 solution in water.



The ppt is filtered, washed, dried and weighed.

The % of S is calculated from mass of BaSO_4 .

Let mass of organic compound taken = W g

mass of BaSO_4 formed = W_1 g

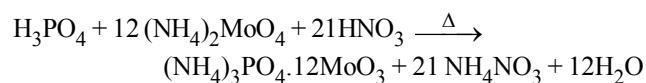
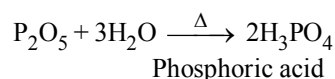
1 mol of BaSO_4 = 233 g BaSO_4 = 32 g S

W_1 g BaSO_4 contains $\frac{32 \times W_1}{233}$ g S

$$\% \text{ of S} = \frac{32 \times W_1 \times 100}{233 \times W}$$

(v) **Phosphorus**

A known mass of an organic compound is heated with fuming HNO_3 . Carbon and hydrogen are oxidised to CO_2 and H_2O respectively while P in compound is oxidised to H_3PO_4 . It is precipitated as ammonium phosphomolybdate, $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$ by adding ammonium molybdate. The precipitate of $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$ is then filtered, washed, dried and weighed.



Let mass of organic compound = W g

mass of ammonium phosphomolybdate = W_1 g

Molar mass of $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$ = 1877 g

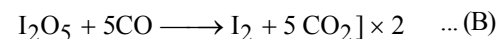
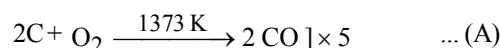
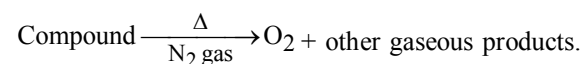
$$\% \text{ of P} = \frac{31 \times W_1 \times 100}{1877 \times W} \%$$

If P is estimated as $\text{Mg}_2\text{P}_2\text{O}_7$.

$$\% \text{ of P} = \frac{62 \times W_1 \times 100}{222 \times W} \%$$

(vi) **Oxygen**

% of O_2 = 100 – (sum of % of all other elements)



On solving (A) and (B), we find that each mole of O_2 liberated produces 2 moles of CO_2 .

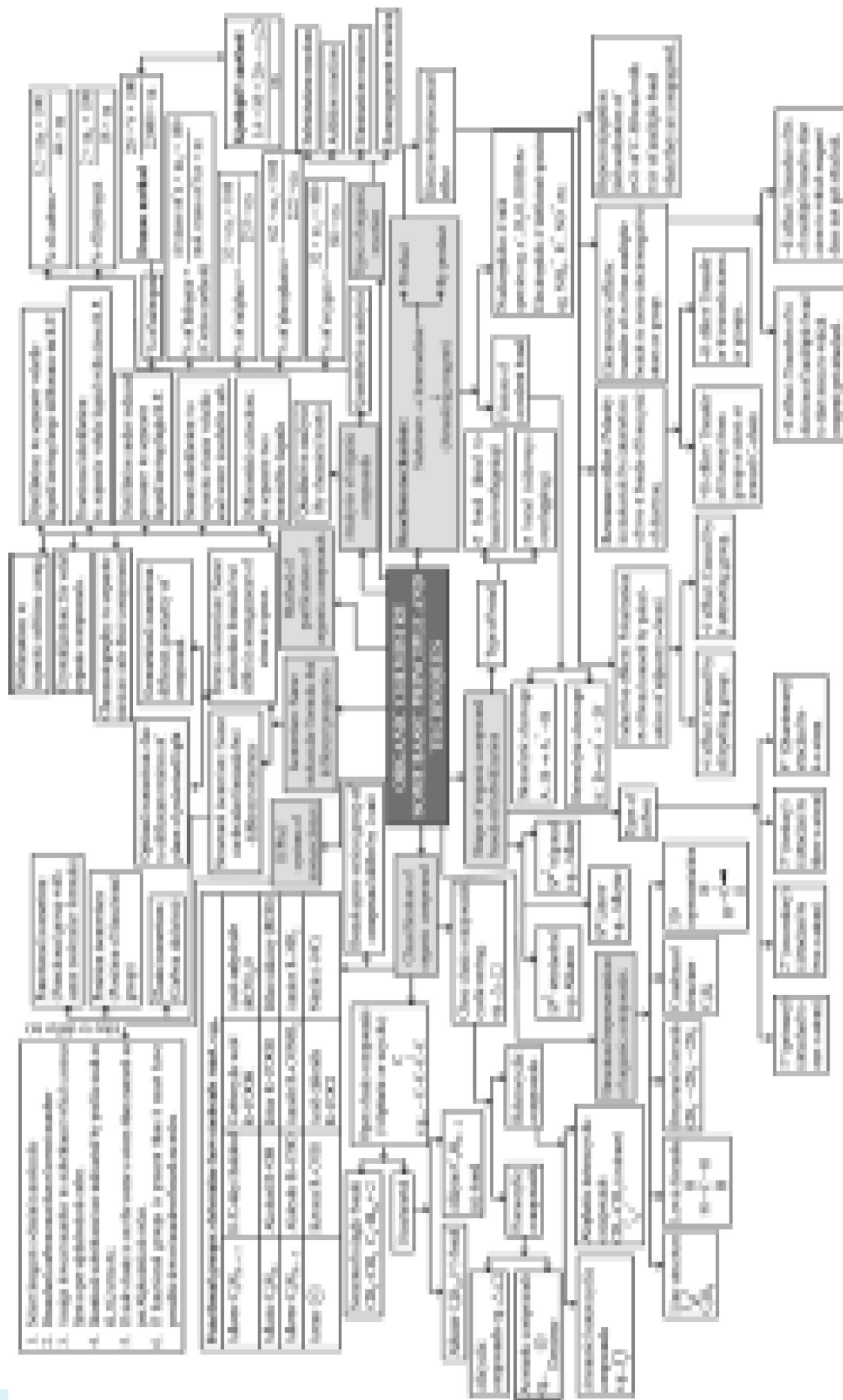
Let mass of organic compound = W g

mass of CO_2 produced = W_1 g

$$\therefore m_1 \text{ g of CO}_2 \text{ is obtained from } \frac{32 \times W_1}{88} \text{ g O}_2$$

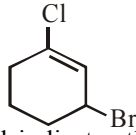
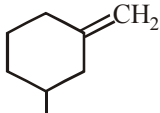
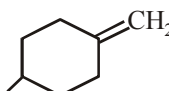
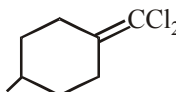
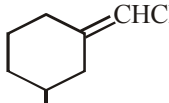
$$\therefore \% \text{ of O}_2 = \frac{32 \times W_1 \times 100}{88 \times W} \%$$

CONCEPT MAP

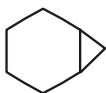


EXERCISE - 1

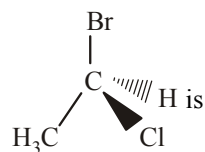
Conceptual Questions

- The number of enantiomers of the compound $\text{CH}_3\text{CHBrCHBrCOOH}$ is :
(a) 2 (b) 3
(c) 4 (d) 6
- The IUPAC name of $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}(\text{CH}_3)_2$ is:
(a) 4-methyl-2-pentyne
(b) 4,4-dimethyl-2-butyne
(c) methyl isopropyl acetylene
(d) 2-methyl-4-pentyne
- Which pairs have same percentage of carbon?
(a) CH_3COOH and $\text{C}_6\text{H}_{12}\text{O}_6$
(b) CH_3COOH and $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
(c) CH_3COOH and $\text{C}_2\text{H}_5\text{OH}$
(d) $\text{C}_6\text{H}_{12}\text{O}_6$ and $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
- The IUPAC name of the compound :
 $\text{CH}_2 - \text{CH} - \text{CH}_2$ is
| | |
CN CN CN
(a) 1, 2, 3-trinitropropane
(b) 1, 2, 3-tricyanopropane
(c) 3-cyanopentane-1, 5-dinitrile
(d) 1, 2, 3-pentanetrinitrile
- The IUPAC name of the compound $\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_2\text{OCH}_2\text{CH}_3$ is :
(a) 3-ethoxy-1-methoxypropane
(b) 1-ethoxy-3-methoxypropane
(c) 2,5-dioxyhexane
(d) ethoxypropane oxymethane
- The absolute configuration of the following :
 $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H} - \text{C} - \text{Cl} \\ | \\ \text{Cl} - \text{C} - \text{H} \\ | \\ \text{C}_2\text{H}_5 \end{array}$ is
(a) 2S, 3R (b) 2S, 3S
(c) 2R, 3S (d) 2R, 3R
- Which of the following compounds has wrong IUPAC name?
(a) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{COO} - \text{CH}_2\text{CH}_3 \rightarrow$ ethyl butanoate
(b) $\text{CH}_3 - \text{CH} - \text{CH}_2 - \text{CHO} \rightarrow$ 3-methyl-butanal
|
 CH_3
(c) $\text{CH}_3 - \text{CH} - \text{CH} - \text{CH}_3 \rightarrow$ 2-methyl-3-butanol
| |
 $\text{OH} \quad \text{CH}_3$
(d) $\text{CH}_3 - \text{CH} - \text{C}(=\text{O}) - \text{CH}_2 - \text{CH}_3$
|
 CH_3
 $\rightarrow$ 2-methyl-3-pentanone
- The IUPAC name of the compound shown below is :
(a) 3-bromo-1-chlorocyclohexene
(b) 1-bromo-3-chlorocyclohexene
(c) 2-bromo-6-chlorocyclohex-1-ene
(d) 6-bromo-2-chlorocyclohexene

- In Lassaigne test a violet colour obtained indicates the presence of
(a) sulphur (b) halogen
(c) nitrogen (d) None of these
- IUPAC name of the compound is :
 $\text{H} - \overset{\text{O}}{\underset{\text{||}}{\text{C}}} - \text{CH}_2 - \text{CH}_2 - \text{OCH}_3$
(a) 2-methoxypropanal (b) methoxypropanal
(c) 3-methoxypropanal (d) 2-formyl methoxyethane
- The number of secondary hydrogens in 2, 2-dimethylbutane is :
(a) 8 (b) 6 (c) 4 (d) 2
- The geometrical isomerism is shown by
(a)  (b) 
(c)  (d) 
- The IUPAC name for
 $\text{CH}_3 - \overset{\text{Cl}}{\underset{\text{H}}{\text{C}}} - \text{CH}_2 - \text{CH} = \text{CH} - \text{CH}_3$ is
(a) 5-chlorohex-2-ene
(b) 2-chlorohex-5-ene
(c) 1-chloro-1-methylpent-3-ene
(d) 5-chloro-5-methylpent-2-ene
- Among the following the most stable compound is :
(a) *cis*-1, 2-cyclohexanediol
(b) *trans*-1, 2-cyclohexanediol
(c) *cis*-1, 3-cyclohexanediol
(d) *trans*-1, 3-cyclohexanediol
- Which of the following compounds contains 1°, 2°, 3° as well as 4° carbon atoms ?
(a) Neopentane (b) 2-methyl pentane
(c) 2,3-dimethyl butane (d) 2,2,3-trimethyl pentane
- Correct configuration of the following is
 $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_3 - \text{C} - \text{OH} \\ | \\ \text{H} \end{array}$
(a) 1S, 2S (b) 1S, 2R (c) 1R, 2R (d) 1R, 2S

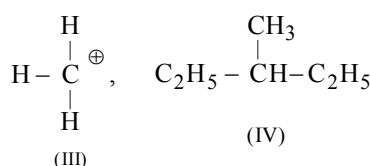
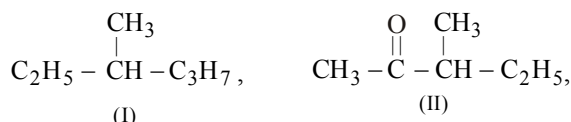
17. The correct name for the following hydrocarbon is
 (a) Tricyclo[4.1.0] heptane
 (b) Bicyclo[5.2.1] heptane
 (c) Bicyclo[4.1.0] heptane
 (d) Bicyclo[4.1.0] hexane
18. The Beilstein test for organic compounds is used to detect
 (a) nitrogen (b) sulphur (c) carbon (d) halogens
19. Which of the following is an optically active compound?
 (a) 1-Butanol (b) 1-Propanol
 (c) 2-Chlorobutane (d) 4-Hydroxyheptane
20. An important chemical method to resolve a racemic mixture makes use of the formation of
 (a) a meso compound (b) enantiomers
 (c) diastereomers (d) racemates
21. Lassaigne's test is used in qualitative analysis to detect
 (a) nitrogen (b) sulphur
 (c) chlorine (d) All of these
22. The process of separation of a racemic modification into *d* and *l*-enantiomers is called
 (a) Resolution (b) Dehydration
 (c) Revolution (d) Dehydrohalogenation
23. Tautomerism will be exhibited by
 (a) $(\text{CH}_3)_2\text{NH}$ (b) $(\text{CH}_3)_3\text{CNO}$
 (c) R_3CNO_2 (d) RCH_2NO_2
24. Lassaigne's test for the detection of nitrogen fails in
 (a) $\text{NH}_2\text{CONHNH}_2 \cdot \text{HCl}$ (b) $\text{NH}_2\text{NH}_2 \cdot \text{HCl}$
 (c) NH_2CONH_2 (d) $\text{C}_6\text{H}_5\text{NHNH}_2 \cdot \text{HCl}$
25. A compound of molecular formula of C_7H_{16} shows optical isomerism, compound will be
 (a) 2,3-Dimethylpentane (b) 2,2-Dimethylpentane
 (c) 2,4-Dimethylpentane (d) None of these
26. $\text{CH}_2=\text{C}(\text{O})-\text{CH}_3$ and $\text{CH}_2=\text{C}(\text{O}^-)-\text{CH}_3$ are
 (a) resonating structures (b) tautomers
 (c) geometrical isomers (d) optical isomers
27. Which of the following pairs of compounds are enantiomers?
- (a) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{HO}-\text{C}-\text{H} \\ | \\ \text{H}-\text{C}-\text{OH} \\ | \\ \text{CH}_3 \end{array}$ and $\begin{array}{c} \text{CH}_3 \\ | \\ \text{HO}-\text{C}-\text{H} \\ | \\ \text{HO}-\text{C}-\text{H} \\ | \\ \text{CH}_3 \end{array}$
- (b) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}-\text{C}-\text{OH} \\ | \\ \text{HO}-\text{C}-\text{H} \\ | \\ \text{CH}_3 \end{array}$ and $\begin{array}{c} \text{CH}_3 \\ | \\ \text{HO}-\text{C}-\text{H} \\ | \\ \text{H}-\text{C}-\text{HO} \\ | \\ \text{CH}_3 \end{array}$
- (c) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}-\text{C}-\text{OH} \\ | \\ \text{HO}-\text{C}-\text{H} \\ | \\ \text{CH}_3 \end{array}$ and $\begin{array}{c} \text{CH}_3 \\ | \\ \text{HO}-\text{C}-\text{H} \\ | \\ \text{HO}-\text{C}-\text{H} \\ | \\ \text{CH}_3 \end{array}$
- (d) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}-\text{C}-\text{OH} \\ | \\ \text{HO}-\text{C}-\text{H} \\ | \\ \text{CH}_3 \end{array}$ and $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}-\text{C}-\text{OH} \\ | \\ \text{H}-\text{C}-\text{OH} \\ | \\ \text{CH}_3 \end{array}$
28. Kjeldahl's method is used in the estimation of
 (a) nitrogen (b) halogens
 (c) sulphur (d) oxygen



29. The chirality of the compound

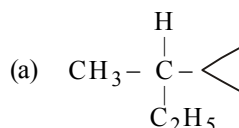


- (a) R (b) S (c) E (d) Z
30. Among the following four structures I to IV,



it is true that

- (a) only I and II are chiral compounds
 (b) only III is a chiral compound
 (c) only II and IV are chiral compounds
 (d) all four are chiral compounds
31. Which of the following is optically inactive?
- (a) $\begin{array}{c} \text{H} \\ | \\ \text{H}_3\text{C}-\text{C}-\text{Cl} \\ | \\ \text{Cl}-\text{C}-\text{CH}_3 \\ | \\ \text{H} \end{array}$ (b) $\begin{array}{c} \text{H} \\ | \\ \text{Cl}-\text{C}-\text{CH}_3 \\ | \\ \text{H}_3\text{C}-\text{C}-\text{Cl} \\ | \\ \text{H} \end{array}$
- (c) $\begin{array}{c} \text{H} \\ | \\ \text{H}_3\text{C}-\text{C}-\text{Cl} \\ | \\ \text{H}_3\text{C}-\text{C}-\text{Cl} \\ | \\ \text{H} \end{array}$ (d) none of these.
32. C_8H_{16} that can form *cis-trans* geometrical isomers and also has a chiral centre, is
- (a) (b)
- (c) Both of these (d) None of these
33. Which of the following will have a mesoisomer also?
 (a) 2,3-Dichloropentane (b) 2,3-Dichlorobutane
 (c) 2-Chlorobutane (d) 2-Hydroxypropanoic acid
34. Amongst the following compounds, the optically active alkane having lowest molecular mass is



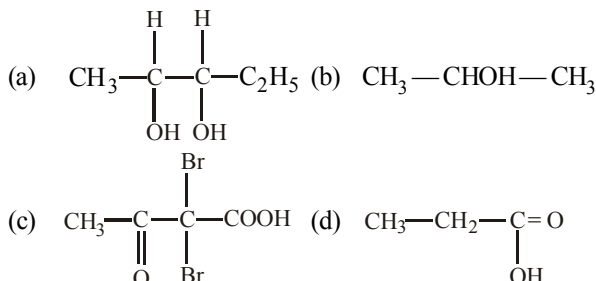
- (b) $\text{CH}_3-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}_3$
 (c) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$
 (d) $\text{CH}_3-\text{CH}_2-\text{C}\equiv\text{CH}$

35. The most suitable method for separation of a 1 : 1 mixture of *ortho* and *para* nitrophenols is
 (a) sublimation (b) chromatography
 (c) crystallization (d) steam distillation

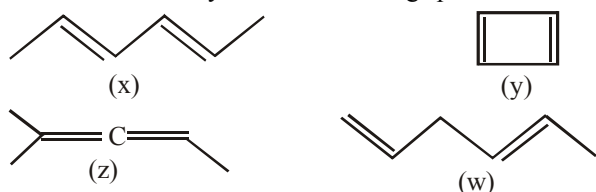
36. Which one of the following is a technique most suitable for purification of cyclohexanone from a mixture containing benzoic acid, isoamyl alcohol, cyclohexane and cyclohexanone?

- (a) Crystallization (b) Sublimation
(c) IR spectroscopy (d) Gas chromatography

37. Which of the following is optically active ?



38. The correct stability order of following species is –



- (a) $x > y > w > z$ (b) $y > x > w > z$
(c) $x > w > z > y$ (d) $z > x > y > w$

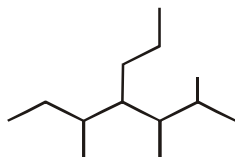
39. An aromatic compound of formula $\text{C}_7\text{H}_7\text{Cl}$ has in all isomers :

- (a) 5 (b) 2 (c) 4 (d) 3

40. How many optically active stereoisomers are possible for lactic acid ?

- (a) 1 (b) 2 (c) 4 (d) 3

41. The correct IUPAC name for



- (a) 5-methyl-4-(1'-2'-demethylpropyl) heptane
(b) 3-methyl-4-(1',2'-dimethylpropyl) heptane
(c) 2,3,5-trimethyl-4-propylheptane
(d) 4-propyl-2,3,5-trimethylpeptane

42. Geometrical isomerism is possible in :

- (a) $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$
(b) $\text{CH}_3\text{CH}=\text{CHCH}_3$
(c) $\text{CH}_3\text{CH}=\text{CH}_2$
(d) $\text{ClH}_2\text{C}-\text{CH}_2\text{Cl}$

43. How many acyclic structural isomers are possible for the compound with molecular formula $\text{C}_3\text{H}_3\text{Cl}_3$?

- (a) 3 (b) 2 (c) 5 (d) 6

44. The number of optical enantiomorphs of tartaric acid is/are:

- (a) 1 (b) 4 (c) 3 (d) 2

45. Which of the cations in more stable ?

- (a) $\text{R}'-\text{C}^+-\text{OR}$ (b) $\text{R}'-\text{C}^+=\text{OR}$
(c) both equal (d) both are unstable

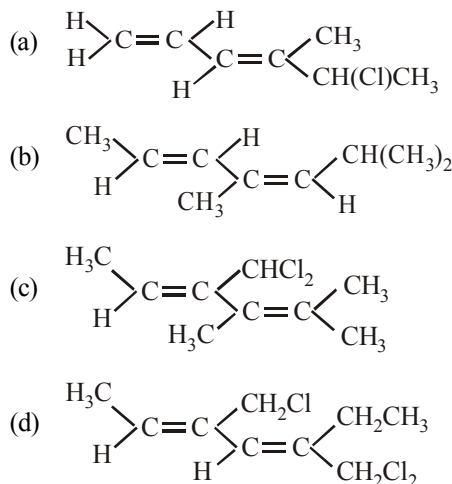
46. Geometrical isomerism is possible in case of :

- (a) pentene-2 (b) propane
(c) pentane (d) ethene

47. The number of isomers of the compound with molecular formula $\text{C}_2\text{H}_2\text{Br}_2$ is

- (a) 4 (b) 3 (c) 5 (d) 2

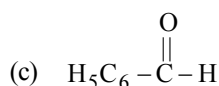
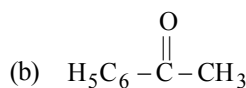
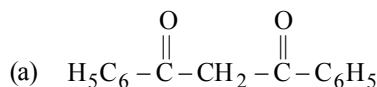
48. One among the following will show optical activity?



49. The number of ether isomers possible for $\text{C}_4\text{H}_{10}\text{O}$ are :

- (a) 2 (b) 5 (c) 4 (d) 3

50. Keto-enol tautomerism is observed in :



(d) Both (a) and (b)

51. The best method for the separation of naphthalene and benzoic acid from their mixture is:

- (a) distillation (b) sublimation
(c) chromatography (d) crystallisation

52. Isomerism due to rotation around C–C single bond is :

- (a) Enantiomerism (b) Position isomerism
(c) Conformation (d) Diastereoisomerism

53. If a compound has n asymmetric carbon atoms, then maximum number of optical isomers are given by the formula

- (a) 2^n (b) $\left(\frac{1}{2}\right)^n$ (c) $\sqrt{2n}$ (d) $2\sqrt{n}$

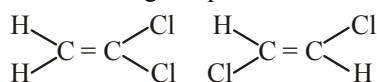
54. Which is a chiral molecule?

- (a) 2,2-dimethylbutanoic acid
(b) 4-methylpentanoic acid
(c) 3-methylpentanoic acid
(d) 3,3-dimethylbutanoic acid

55. In Kjeldahl's method, CuSO_4 acts as

- (a) oxidising agent (b) reducing agent
(c) hydrolysing agent (d) catalytic agent

56. The following compounds differ in

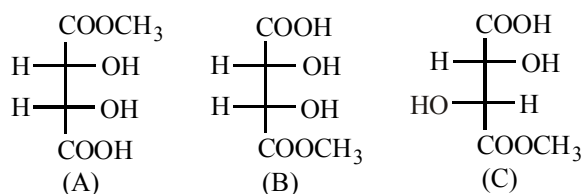


- (a) configuration (b) conformation
(c) structure (d) chirality

57. A mixture of methyl alcohol and acetone can be separated by

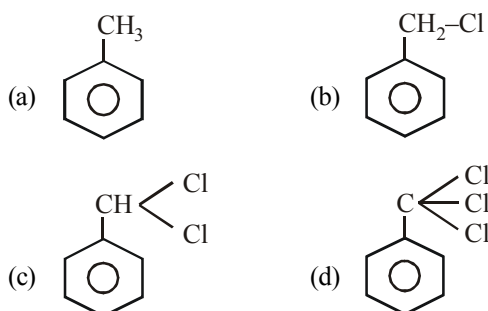
- (a) distillation
(b) fractional distillation
(c) steam distillation
(d) distillation under reduced pressure

58. The correct statement about the compounds (A), (B) and (C) is :

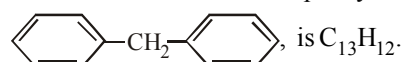


- (a) (A) and (B) are identical
(b) (A) and (B) are diastereomers
(c) (A) and (C) are enantiomers
(d) (A) and (B) are enantiomers

59. Which compound on nitration will give highest amount of m-substituted product ?



60. The molecular formula of diphenyl methane,



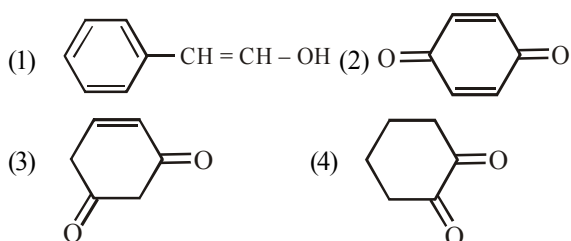
is $C_{13}H_{12}$. How many structural isomers are possible when one of the hydrogens is replaced by a chlorine atom?

- (a) 6 (b) 4 (c) 8 (d) 7

61. Nitroethane can exhibit one of the following kind of isomerism:

- (a) metamerism (b) optical activity
(c) tautomerism (d) position isomerism

62. Tautomerism is exhibited by –



- (a) (1), (3) and (4) (b) (2), (3), and (4)
(c) All of these (d) None of these

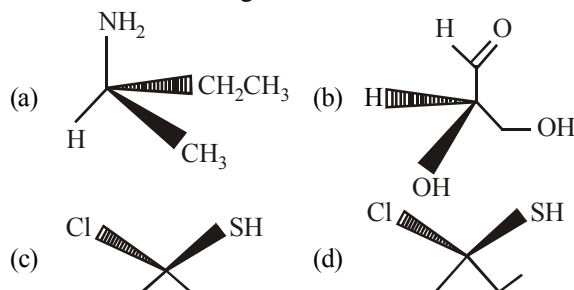
63. Which of the substance is purified by sublimation?

- (a) Benzoic acid (b) Camphor
(c) Naphthalene (d) All of these

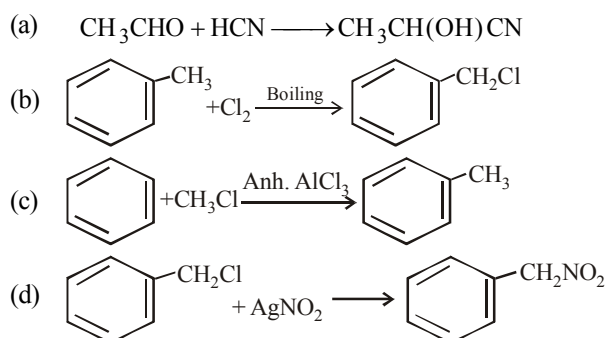
64. The most reactive compound for electrophilic nitration is

- (a) Benzene (b) Nitrobenzene
(c) Benzoic acid (d) Toluene.

65. Which of the following molecules is achiral?



66. Which one of the following is a free-radical substitution reaction?



67. Which of the following undergoes nucleophilic substitution exclusively by S_N1 mechanism?

- (a) Ethyl chloride (b) Isopropyl chloride
(c) Chlorobenzene (d) Benzyl chloride

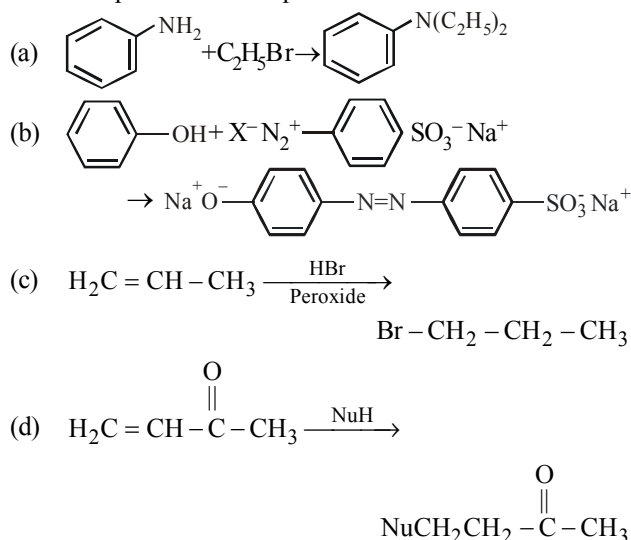
68. Styrene can be purified by

- (a) Simple distillation (b) Fractional distillation
(c) Steam distillation (d) Vacuum distillation

69. The correct order regarding the electronegativity of hybrid orbitals of carbon is

- (a) $sp > sp^2 > sp^3$ (b) $sp < sp^2 > sp^3$
(c) $sp < sp^2 < sp^3$ (d) $sp > sp^2 < sp^3$

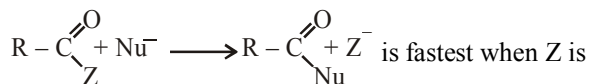
70. The example of an electrophilic substitution reaction is :



71. Which one of the following does not have sp^2 hybridized carbon ?

- (a) Acetonitrile (b) Acetic acid
(c) Acetone (d) Acetamide

72. Rate of the reaction



- (a) OC_2H_5 (b) NH_2 (c) Cl (d) $OCOCH_3$

73. The increasing order of stability of the following free radicals is

- (a) $(C_6H_5)_2\dot{C}H < (C_6H_5)_3\dot{C} < (CH_3)_3\dot{C} < (CH_3)_2\dot{C}H$
(b) $(CH_3)_2\dot{C}H < (CH_3)_3\dot{C} < (C_6H_5)_2\dot{C}H < (C_6H_5)_3\dot{C}$
(c) $(CH_3)_2\dot{C}H < (CH_3)_3\dot{C} < (C_6H_5)_2\dot{C}H < (C_6H_5)_3\dot{C}$
(d) $(C_6H_5)_3\dot{C} < (C_6H_5)_2\dot{C}H < (CH_3)_3\dot{C} < (CH_3)_2\dot{C}H$

74. Steam distillation is used for the purification of :

- (a) covalent compounds
(b) electrovalent compounds
(c) co-ordinate compounds
(d) volatile compounds

75. Which of the following is strongest nucleophile –

- (a) Br^- (b) $:OH^-$
(c) $:CN^-$ (d) $C_2H_5O^-$

76. Strongest acid among the following is :

- (a) CCl_3COOH (b) CH_3COOH
(c) CF_3COOH (d) CBr_3COOH

77. The prussian blue colour obtained in the Lassaigne's test for nitrogen is due to the formation of

- (a) Iron (II) hexacyanoferrate (III)
(b) Iron (III) hexacyanoferrate (II)
(c) Iron (III) hexacyanoferrate (III)
(d) Iron (II) hexacyanoferrate (II)

78. Which one of the following species is isoelectronic with ammonia ?

- (a) CH_2^- (b) $^+CH_2$ (c) $^-CH_3$ (d) $^+CH_3$

79. A *meta*-directing functional group is :

- (a) $-COOH$ (b) OH
(c) $-CH_3$ (d) $-Br$

80. The following compound will undergo electrophilic substitution more readily than benzene :

- (a) Nitrobenzene (b) Benzoic acid
(c) Benzaldehyde (d) Phenol

81. In Kjeldahl's method for the estimation of nitrogen, the formula used is

- (a) $\%N = \frac{1.4VW}{N}$ (b) $\%N = \frac{1.4VW}{V}$
(c) $\%N = \frac{VNW}{1.8}$ (d) $\%N = \frac{1.4VN}{W}$

82. The number of σ and π bonds in a molecule of acetonitrile are respectively

- (a) 2, 5 (b) 3, 4 (c) 4, 3 (d) 5, 2

83. An asymmetric carbon atom is that in which :

- (a) three similar and one dissimilar groups
(b) two similar and two dissimilar groups
(c) all the different groups
(d) none of the above

84. Which of the following pairs represent electrophiles?

- (a) $AlCl_3, H_2O$ (b) SO_3, NO_2^+
(c) BF_3, H_2O (d) NH_3, SO_3

85. Chromatography is a valuable method for the separation, isolation, purification and identification of the constituents of a mixture and it is based on general principle of

- (a) phase rule (b) phase distribution
(c) interphase separation (d) phase operation

86. The shape of methyl carbanion is similar to that of –

- (a) BF_3 (b) NH_3
(c) methyl free radical (d) methyl carbocation

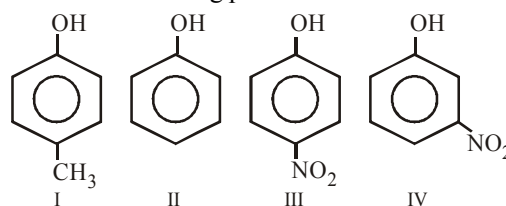
87. The replacement of chlorine of chlorobenzene to give phenol requires drastic conditions, but the chlorine of 2,4-dinitrochlorobenzene is readily replaced since,

- (a) nitro groups make the aromatic ring electron rich at ortho/para positions
(b) nitro groups withdraw electrons from the meta position of the aromatic ring
(c) nitro groups donate electrons at meta position
(d) nitro groups withdraw electrons from ortho/para positions of the aromatic ring

88. The optical inactivity due to internal compensation can be exhibited by :

- (a) dextro-tartaric acid (b) laevo-tartaric acid
(c) racemic-tartaric acid (d) meso-tartaric acid

89. Consider the following phenols :



The decreasing order of acidity of the above phenols is

- (a) $III > IV > II > I$ (b) $II > I > IV > III$
(c) $I > IV > II > III$ (d) $III > IV > I > II$

90. Absolute alcohol cannot be obtained by simple fractional distillation because

- (a) Pure C_2H_5OH is unstable
(b) C_2H_5OH form hydrogen bonds with water
(c) Boiling point of C_2H_5OH is very close to that of water
(d) Constant boiling azeotropic mixture is formed with water

91. Impure glycerine can be purified by

- (a) Steam distillation (b) Simple distillation
(c) Vacuum distillation (d) Extraction with a solvent

92. Consider the acidity of the carboxylic acids :

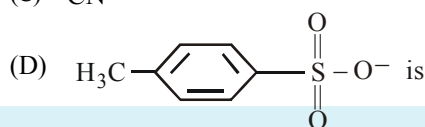
- (A) $PhCOOH$ (B) $o-NO_2C_6H_4COOH$
(C) $p-NO_2C_6H_4COOH$ (D) $m-NO_2C_6H_4COOH$

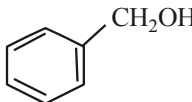
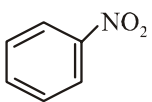
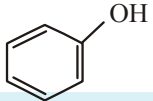
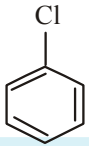
Which of the following order is correct ?

- (a) $B > D > A > C$ (b) $B > D > C > A$
(c) $A > B > C > D$ (d) $B > C > D > A$

93. The decreasing order of nucleophilicity among the nucleophiles

- (A) $CH_3C(=O)O^-$ (B) CH_3O^-
(C) CN^-

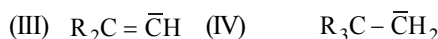
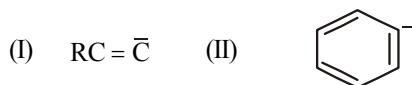


- (a) (C), (B), (A), (D) (b) (B), (C), (A), (D)
(c) (D), (C), (B), (A) (d) (A), (B), (C), (D)
94. 1,2-Dihydroxybutane has the formula
(a) C(OH)-C(OH)-C-C (b) $\text{C-C(OH)}_2\text{-C-C}$
(c) (OH)C-C-C-C(OH) (d) C-C(OH)-C(OH)-C
95. Which of the following represents the correct order of stability of the given carbocations ?
(a) $\text{F}_3\text{C}^+ > \text{F}_3\text{C}-\text{C}^+ > \text{CH}_3^+$ (b) $\text{H}_3\text{C}^+ > \text{F}_3\text{C}-\text{C}^+ > \text{F}_3\text{C}^+$
(c) $\text{F}_3\text{C}-\text{C}^+ > \text{F}_3\text{C}^+ > \text{H}_3\text{C}^+$ (d) $\text{F}_3\text{C}-\text{C}^+ > \text{H}_3\text{C}^+ > \text{F}_3\text{C}^+$
96. A bottle containing two immiscible liquids is given to you. They may be separated by using a
(a) Fractionating column (b) Separating funnel
(c) Vacuum distillation (d) Steam distillation
97. Distillation is used to separate liquids which differ in the b.p by
(a) 5°C (b) 10°C
(c) $30-80^\circ\text{C}$ (d) 100°C
98. Example of chain isomerism are :
(a) Ortho and meta toluic acids
(b) Methyl acetate and ethyl formate
(c) Pentanoic acid and 2-methyl butanoic acid
(d) 2-pentanone and 3-pentanone
99. $\text{CH}_3-\text{CHCl}-\text{CH}_2-\text{CH}_3$ has a chiral centre. which one of the following represents its R-configuration?
(a) $\begin{array}{c} \text{C}_2\text{H}_5 \\ | \\ \text{H}-\text{C}-\text{CH}_3 \\ | \\ \text{Cl} \end{array}$ (b) $\begin{array}{c} \text{C}_2\text{H}_5 \\ | \\ \text{Cl}-\text{C}-\text{CH}_3 \\ | \\ \text{H} \end{array}$
(c) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}-\text{C}-\text{Cl} \\ | \\ \text{C}_2\text{H}_5 \end{array}$ (d) $\begin{array}{c} \text{C}_2\text{H}_5 \\ | \\ \text{H}_3\text{C}-\text{C}-\text{Cl} \\ | \\ \text{H} \end{array}$
100. For (i) I^- , (ii) Cl^- , (iii) Br^- , the increasing order of nucleophilicity would be
(a) $\text{Cl}^- < \text{Br}^- < \text{I}^-$ (b) $\text{I}^- < \text{Cl}^- < \text{Br}^-$
(c) $\text{Br}^- < \text{Cl}^- < \text{I}^-$ (d) $\text{I}^- < \text{Br}^- < \text{Cl}^-$
101. The order of decreasing reactivity towards an electrophilic reagent, for the following would be
(i) benzene (ii) toluene
(iii) chlorobenzene (iv) phenol
(a) (ii) > (iv) > (i) > (iii) (b) (iv) > (iii) > (ii) > (i)
(c) (iv) > (ii) > (i) > (iii) (d) (i) > (ii) > (iii) > (iv)
102. Which one of the following is most reactive towards electrophilic attack ?
(a)  (b) 
(c)  (d) 

103. How many stereoisomers does this molecule have ?
 $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CHBrCH}_3$

(a) 4 (b) 6 (c) 8 (d) 2

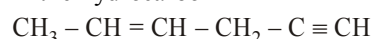
104. The stability of carbanions in the following :



is in the order of :

(a) (I) > (II) > (III) > (IV) (b) (II) > (III) > (IV) > (I)
(c) (IV) > (II) > (III) > (I) (d) (I) > (III) > (II) > (IV)

105. In the hydrocarbon

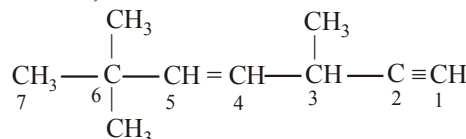


6 5 4 3 2 1

The state of hybridization of carbons 1, 3 and 5 are in the following sequence :

(a) sp^2, sp, sp^3 (b) sp, sp^3, sp^2
(c) sp, sp^2, sp^3 (d) sp^3, sp^2, sp

106. The state of hybridization of C_2 , C_3 , C_5 and C_6 of the hydrocarbon,



is in the following sequence:

(a) sp^3, sp^2, sp^2 and sp (b) sp, sp^2, sp^2 and sp^3
(c) sp, sp^2, sp^3 and sp^2 (d) sp, sp^3, sp^2 and sp^3

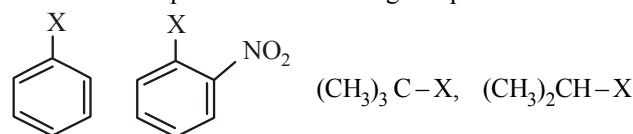
107. Which of the following compounds will exhibit cis-trans (geometrical) isomerism?

(a) Butanol (b) 2-Butyne
(c) 2-Butenol (d) 2-Butene

108. Cyclohexanol (I), acetic acid (II), 2, 4, 6-trinitrophenol (III) and phenol (IV) are given. In these the order of decreasing acidic character will be :

(a) III > II > IV > I (b) II > III > I > IV
(c) II > III > IV > I (d) III > IV > II > I

109. The correct order of increasing reactivity of $\text{C}-\text{X}$ bond towards nucleophile in the following compounds is:

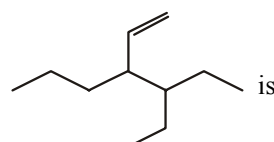


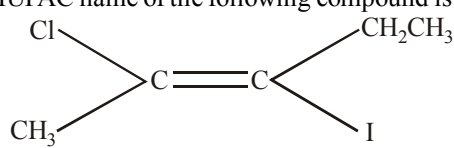
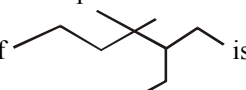
(I) (II) (III) (IV)
(a) I < II < IV < III (b) II < III < I < IV
(c) IV < III < I < II (d) III < II < I < IV

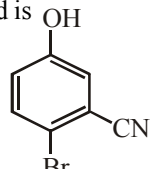
110. Among the given compounds, the most susceptible to nucleophilic attack at the carbonyl group is:

(a) $\text{CH}_3\text{COOCH}_3$ (b) CH_3CONH_2
(c) $\text{CH}_3\text{COOCOCH}_3$ (d) CH_3COCl

111. The correct IUPAC name of the compound



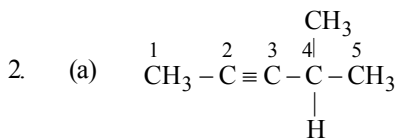
- (a) 4-Ethyl-3-propyl hex-1-ene
(b) 3-Ethyl-4-ethenyl heptane
(c) 3-Ethyl-4-propyl hex-1-ene
(d) 3-(1-ethylpropyl) hex-1-ene
112. The IUPAC name of the following compound is
- 
- (a) trans-2-chloro-3-iodo-2-pentene
(b) cis-3-iodo-4-chloro-3-pentene
(c) trans-3-iodo-4-chloro-3-pentene
(d) cis-2-chloro-3-iodo-2-pentene
113. The IUPAC name of  is
- (a) 3-ethyl-4, 4-dimethylheptane
(b) 1, 1-diethyl-2,2-dimethylpentane
(c) 4, 4-dimethyl-5,5-diethylpentane
(d) 5, 5-diethyl-4,4-dimethylpentane.
114. Presence of a nitro group in a benzene ring
- (a) deactivates the ring towards electrophilic substitution
(b) activates the ring towards electrophilic substitution
(c) renders the ring basic
(d) deactivates the ring towards nucleophilic substitution.
115. The correct decreasing order of priority for the functional groups of organic compounds in the IUPAC system of nomenclature is
- (a) $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$, $-\text{CHO}$
(b) $-\text{SO}_3\text{H}$, $-\text{COOH}$, $-\text{CONH}_2$, $-\text{CHO}$
(c) $-\text{CHO}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$
(d) $-\text{CONH}_2$, $-\text{CHO}$, $-\text{SO}_3\text{H}$, $-\text{COOH}$
116. The IUPAC name of neopentane is
- (a) 2, 2 dimethylpropane (b) 2 methylpropane
(c) 2, 2 dimethylbutane (d) 2- methylbutane

117. The alkene that exhibits geometrical isomerism is :
- (a) 2- methyl propene (b) 2-butene
(c) 2- methyl -2- butene (d) propene
118. The number of stereoisomers possible for a compound of the molecular formula $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}(\text{OH})-\text{Me}$ is:
- (b) 2 (c) 4 (d) 6 (d) 3
119. Out of the following, the alkene that exhibits optical isomerism is
- (a) 3-methyl-2-pentene (b) 4-methyl-1-pentene
(c) 3-methyl-1-pentene (d) 2-methyl-2-pentene
120. The IUPAC name of the following compound is
- 
- (a) 4-Bromo-3-cyanophenol
(b) 2-Bromo-5-hydroxybenzonitrile
(c) 2- Cyano-4-hydroxybromobenzene
(d) 6-Bromo-3-hydroxybenzonitrile
121. Hyperconjugation involves overlap of the following orbitals
- (a) $\sigma-\sigma$ (b) $\sigma-\pi$ (c) p- p (d) $\pi-\pi$
122. In allene (C_3H_4), the type(s) of hybridisation of the carbon atoms is (are) :
- (a) sp and sp^3 (b) sp and sp^2
(c) only sp^3 (d) sp^2 and sp^3
123. Which one of the following conformations of cyclohexane is chiral?
- (a) Boat (b) Twist boat (c) Rigid (d) Chair.
124. Directing influence of the methyl group can be explained on the basis of
- (a) inductive effect (b) electromeric effect
(c) resonance effect (d) no-bond resonance
125. The number of isomers of the compound C_2BrFCl is
- (a) 3 (b) 4 (c) 5 (d) 6

Hints & Solutions

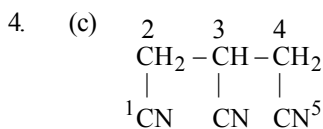
EXERCISE - 1

1. (c) No. of asymmetric carbon = 2
No. of enantiomers = $2^2 = 4$.

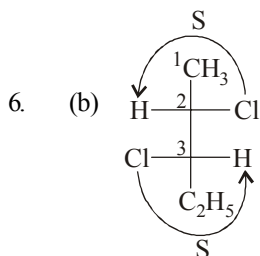
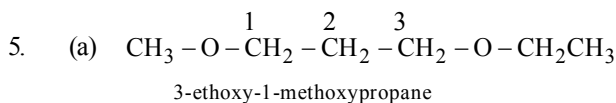


4-methyl-2-pentyne

3. (a) is the answer.
Molecular formula of $\text{CH}_3\text{COOH} = \text{C}_2\text{H}_4\text{O}_2$
Both the compounds given have same empirical formula that is CH_2O . So, percentage of carbon in both of them will be same.

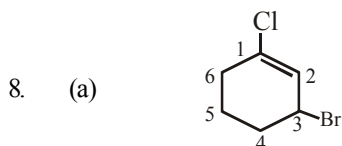


IUPAC name is 3-cyanopentane-1,5-dinitrile



So compound is (2S, 3S)

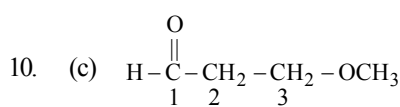
7. (c) The correct name is 3-methylbutan-2-ol



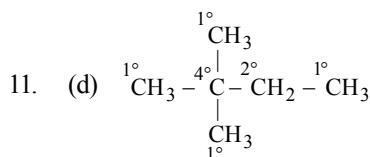
3-bromo-1-chlorocyclohexene

9. (a) Violet colour in Lassaigne's test indicates the presence of sulphur.
$$\text{Na}_2\text{S} + \text{Na}_2[\text{Fe}(\text{NO})(\text{CN})_5] \longrightarrow$$

sodium nitroprusside



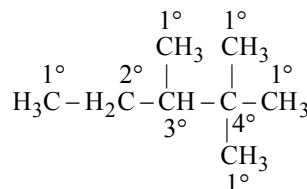
3-methoxypropanal



Thus number of secondary hydrogens is two.

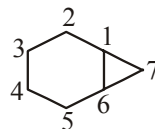
12. (d) Geometrical isomerism is observed when different groups are attached to each of the doubly bonded carbon atom.
13. (a) The IUPAC name of the given compound is 5-chlorohex-2-ene.
14. (d) Among 1, 2- and 1, 3- configurations, 1, 3- is more stable due to less repulsion.
Further among *cis* and *trans* isomers, *trans* is more stable due to less crowding.

15. (d) 2, 2, 3-trimethyl pentane



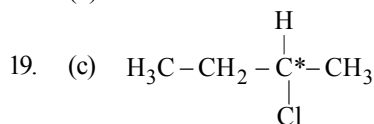
16. (a)

17. (c)



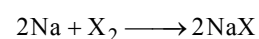
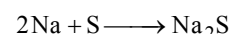
Bicyclo [4, 1, 0] heptane

18. (d)



The compound containing a chiral carbon atom i.e., (a carbon atom which is attached to four different atoms is known as a chiral carbon atom) is optically active. As 2-chlorobutane contains a chiral C* atom hence it is optically active.

20. (c) Diastereomers since they have different melting points, boiling points, solubilities etc.
21. (d) Nitrogen, sulphur and halogens are tested in an organic compound by Lassaigne's test. The organic compound is fused with sodium metal as to convert these elements into ionisable inorganic substances,

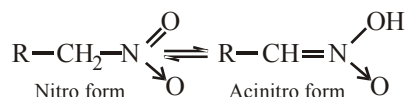


The cyanide, sulphide or halide ions can be confirmed in aqueous solution by usual test.

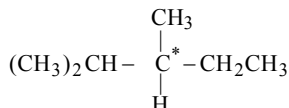
22. (a) Resolution.

23. (d) Tautomerism is exhibited by the oscillation of hydrogen atom between two polyvalent atoms present in the molecule. As (d) has α -hydrogen atom therefore it

shows tautomerism whereas other structures do not.



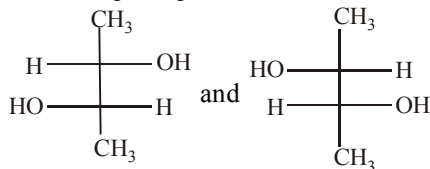
24. (b) Hydrazine (NH_2NH_2) does not contain carbon and hence on fusion with Na metal, it cannot form NaCN; consequently hydrazine does not show Lassaigne's test for nitrogen.
25. (a) A compound is said to exhibit optical isomerism if it atleast contains one chiral carbon atom, which is an atom bonded to 4 different atoms or groups.



26. (a) $\text{CH}_2=\text{C}(\text{O}^-)-\text{CH}_3 \leftrightarrow \text{CH}_2=\text{C}(\text{O}^-)-\text{CH}_3$
- I II

both are resonating structures.

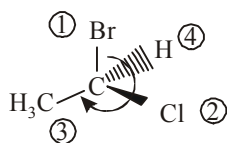
27. (b) Compound which are mirror image of each other and are non superimposable are termed as enantiomers.



These are enantiomers

28. (a) Kjeldahl's method is suitable for estimating nitrogen in those compounds in which nitrogen is linked to carbon and hydrogen. The method is not used in case of nitro, azo and azoxy compound. This method is basically used for estimating nitrogen in food fertilizers and agricultural products.

29. (a) Clockwise rotation.



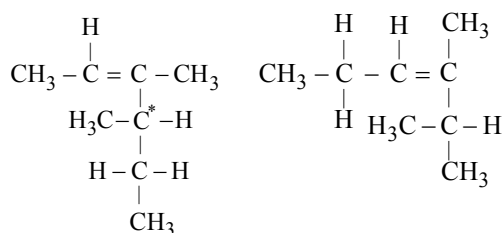
Hence configuration is R.

If the eye travel in a clockwise direction, the configuration is R as the order of priority is $\text{Br} > \text{Cl} > \text{CH}_3 > \text{H}$

30. (a) Carbon atom is connected with four different groups in chiral structure.

31. (c) It has centre of symmetry, so optically inactive.

32. (a) Redrawing the given structures we get,

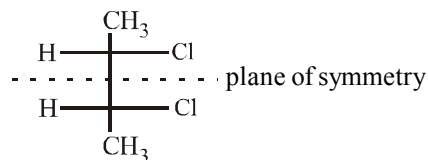


(a)

(b)

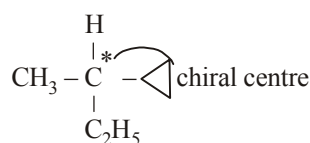
From the above structure, it is evident than structure (a) will show *cis-trans* isomerism and it also has a chiral carbon. On the other hand, structure (b) although shows *cis-trans* isomerism, it has no chiral carbon.

33. (b) The compound has two similar assymmetric C-atoms. It has plane of symmetry and exist in Meso form.



Meso - 2, 3 dichlorobutane

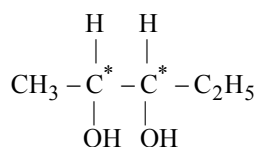
34. (a) Only 2-cyclopropyl butane has a chiral centre.



35. (d) The boiling point of *o*-nitrophenol is less than *para*-nitrophenol due to presence of intramolecular hydrogen bonding. Since *p*-nitrophenol is less volatile in steam than *o*-nitrophenol hence they can be separated by steam distillation.

36. (c) IR spectroscopy is used for the purification of cyclohexanone from a mixture of benzoic acid, isoamyl alcohol, cyclohexane and cyclohexanone. because in this method, each functional group appear at a certain peak. So, cyclohexane can be identified by carbonyl peak.

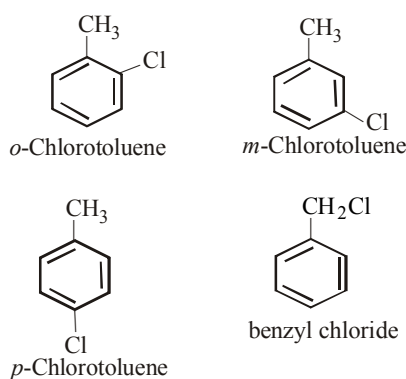
37. (a) Optically active compound has asymmetric carbon atom



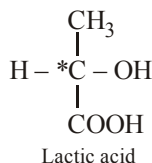
It has two asymmetric carbon (marked as *) atoms and is optically active.

38. (c) x is a conjugated diene system, w is an isolated diene system, z is a cumulated diene system, y is antiaromatic system.

39. (c) $\text{C}_7\text{H}_7\text{Cl}$ has 4 isomers

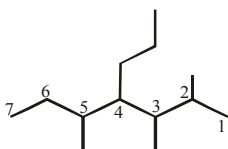


40. (b) No. of optical isomers = 2^x (where x = No. of asymmetric atoms)



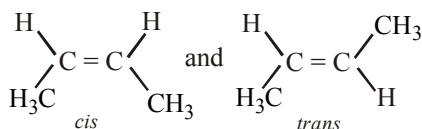
$\therefore$ No. of optical isomers of lactic acid = $(2)^1 = 2$

41. (c) In case two or more chains are of equal length, then the chain with greater number of side chains is selected as the principal chain.

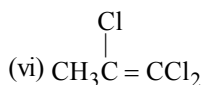
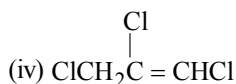
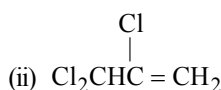


2, 3, 5-Trimethyl-4-propylheptane

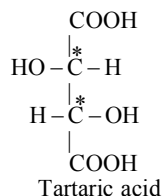
42. (b) Geometrical isomerism is shown by compounds which have $\text{C}=\text{C}$ and two groups attached to same C atoms are different. Choice (b) fulfills both conditions.



43. (d) The parent hydrocarbon of $\text{C}_3\text{H}_3\text{Cl}_3$ is C_3H_6 , i.e. $^3\text{CH}_3\text{CH}=\text{CH}_2$. Three chlorine atoms can be present in following ways.



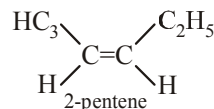
44. (d)



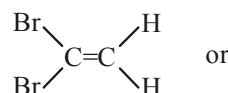
Tartaric acid is an example of a symmetrical molecule, having two asymmetric carbon atoms. In such cases Number of optical enantiomorphs (d and l isomers) = $2^{(n-1)} = 2^1 = 2$

45. (b) Although (b) has positive charge on O (an electronegative element), it is more stable because here every atom has octet of electrons.

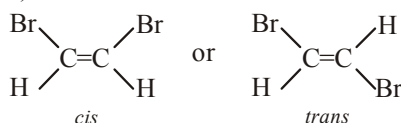
46. (a) Geometrical isomerism is shown by compounds having $\text{C}=\text{C}$ and the two groups attached to each of carbon atom are different



47. (b) The parent hydrocarbon is C_2H_4 or $\text{CH}_2=\text{CH}_2$, so, $\text{C}_2\text{H}_2\text{Br}_2$ can be obtained by replacing two H by two Br for which there are three possibilities.

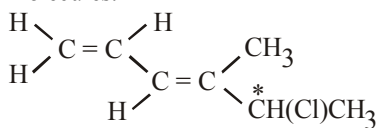


1, 1-dibromoethene



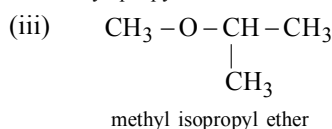
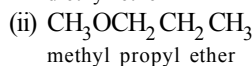
1, 2-dibromoethene

48. (a) Optical activity is the result of chirality (which comes from the presence of chiral carbon in molecule) or the non-superimposability of certain molecules on their mirror images. So, the compound given in option (a) which contains a chiral carbon atom(*) is the only chiral molecule and hence is optically active among the given molecules.

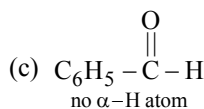
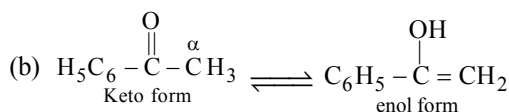
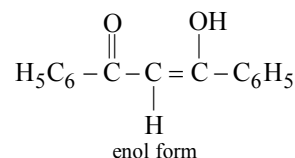
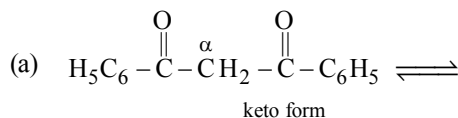


* marked carbon atom is chiral.

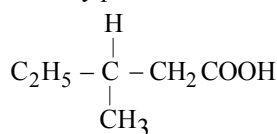
49. (d) (i) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
diethyl ether



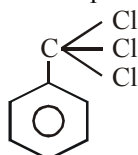
50. (d) Keto-enol tautomerism is shown by carbonyl compounds having α -hydrogen atom.



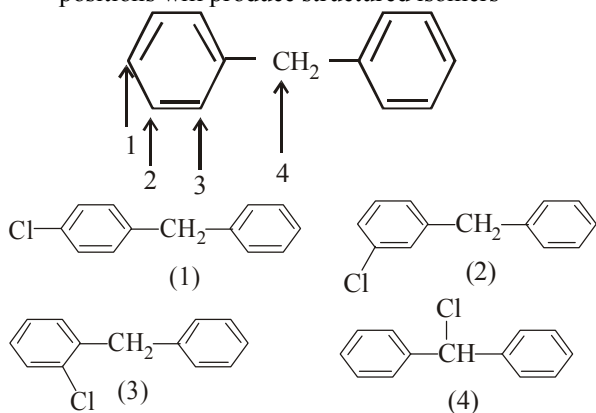
51. (b) Among the given compounds naphthalene is volatile but benzoic acid is non-volatile (it forms a dimer). So, the best method for their separation is sublimation, which is applicable to compounds which can be converted directly into the vapour phase from its solid state on heating and back to the solid state on cooling. Hence it is the most appropriate method.
52. (c) The free rotation across C – C gives rise to many spatial arrangements. These arrangements are called conformers and property is called conformation.
53. (a) Maximum number optical isomers of a compound having n asymmetric carbon atoms is 2^n .
54. (c) Chiral molecules are those molecules which have atleast one asymmetric carbon atom (a carbon atom attached to 4 different groups). This is true in case of 3-methylpentanoic acid.



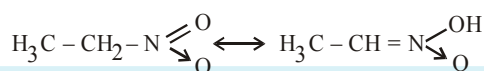
55. (d) Kjeldahl's method depends upon the fact that most of the organic compounds containing nitrogen are quantitatively decomposed to give $(\text{NH}_4)_2\text{SO}_4$ when heated strongly with conc. H_2SO_4 . In this method CuSO_4 acts as catalytic agent.
56. (a) Given structures are representing *cis-trans* isomerism (geometrical) hence differ only in configuration.
57. (a)
58. (d) A and B are enantiomers.
59. (d) $-\text{CH}_3$ group has electron donating inductive effect which decreases with replacement of each hydrogen atom by chlorine atom. Hence highest amount of *m*-substituted product will be obtained on nitration of



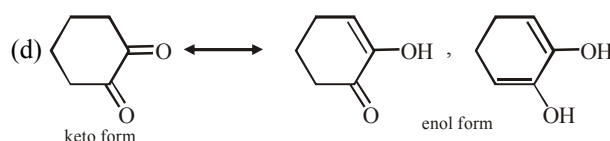
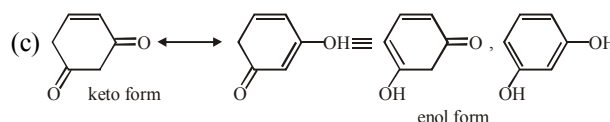
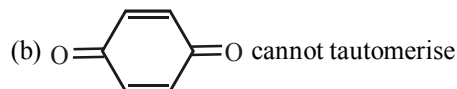
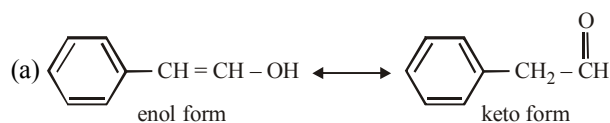
60. (b) In diphenylmethane monochlorination at following positions will produce structured isomers



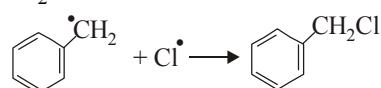
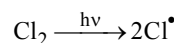
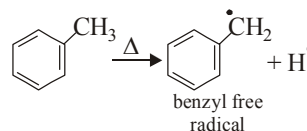
61. (c) Nitroethane ($\text{C}_2\text{H}_5\text{NO}_2$) can exhibit tautomerism among the given kinds of isomerism due to the presence of α -hydrogen which is labile and form nitrolic acid.



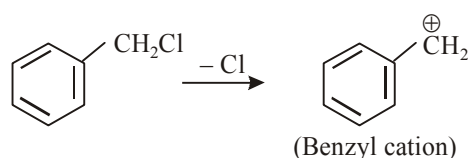
62. (a).



63. (d) Benzoic acid, camphor and naphthalene sublime on heating hence, they are purified by sublimation method.
64. (d) Due to + I-effect of the CH_3 group, toluene has much higher electron density in the ring than benzene, nitrobenzene and benzoic acid as they show – I-effect and hence is most reactive towards nitration.
65. (c) Compound given in option (c) is the only compound which do not have any chiral carbon atom. Thus, it is achiral i.e. do not show optical activity.
66. (b) In the presence of UV rays or energy, by boiling chlorine, free radical is generated which attack the methyl carbon atom of the toluene.



67. (d) $\text{S}_{\text{N}}1$ reactions involve the formation of carbocation, more is stability of carbocation, higher will be the probability of alkyl halide to undergo $\text{S}_{\text{N}}1$ mechanism. Thus decreasing order of $\text{S}_{\text{N}}1$ mechanisms follows order Benzyl > allyl > tertiary alkyl halide > secondary alkyl halide > primary alkyl halides

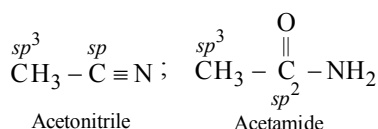
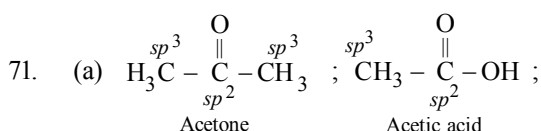


Obtained from $\text{S}_{\text{N}}1$ path. This molecule is resonance stabilised.

68. (d) Styrene can be purified by vacuum distillation.
69. (a) Among the three given hybrid orbitals, sp hybrid orbital is most electronegative. Contribution of s in sp hybrid orbital is maximum so this orbital is closer to nucleus. Naturally it will have greater tendency to pull electron

towards it. Hence it becomes more electronegative and sp^3 becomes least electronegative.

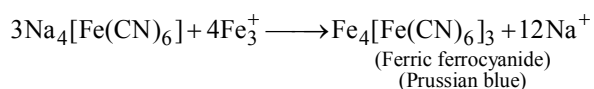
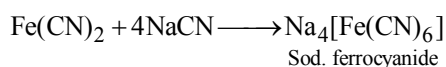
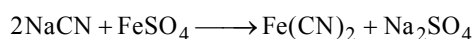
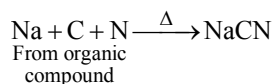
70. (b) It is a coupling reaction of diazonium salt. In it the aromatic ring undergoing attack by diazonium ion must contain a powerful electron releasing group, generally $-OH$, $-NR_2$, $-NHR$ or $-NH_2$. Substitution usually secures para- to the activating group. It has been established experimentally that coupling is electrophilic aromatic substitutions.



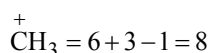
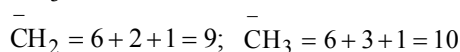
72. (c) Cl^- is the best leaving group among the given option.
73. (b) The order of stability of free radicals
 $(C_6H_5)_3\dot{C} > (C_6H_5)_2\dot{C}H > (CH_3)_3\dot{C} > (CH_3)_2\dot{C}H$
The stabilisation of first two is due to resonance and last two is due to inductive effect.

74. (d) Steam distillation is used for the purification of volatile compounds.
75. (c) The strength of nucleophile depends upon the nature of alkyl group R on which nucleophile has to attack and also on the nature of solvent. The order of strength of nucleophiles follows the order :
 $CN^- > I^- > C_6H_5O^- > OH^- > Br^- > Cl^-$

76. (c) CF_3COOH has highest acidity due to $-I$ effect of F.
77. (b) The chemistry of the Lassaigne's test of nitrogen is



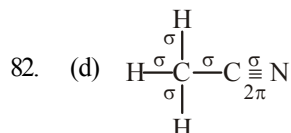
78. (c) Isoelectronic species has same number of electrons. No. of electrons in different species



79. (a) In present case, $\begin{array}{c} O \\ || \\ -C-OH \end{array}$ is the only group whose key atom (atom attached to benzene carbon) is having a multiple bond (one of the characteristics of m -directing groups).

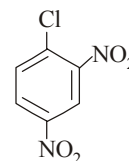
80. (d) Due to +M effect of the $-OH$ group in phenol, electron density in the ortho and para positions of the phenol nucleus increases hence phenol will undergo electrophilic substitution easily than benzene. The other three compounds have electron withdrawing groups, hence they will undergo electrophilic substitution with a difficulty than benzene.

81. (d)



Therefore, the number of σ and π bonds in acetonitrile are 5 and 2 respectively.

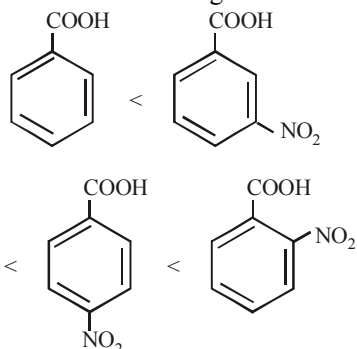
83. (c) An asymmetric carbon atom is that in which all different groups are attached with carbon. Option (c) is correct.
84. (b) Electrophiles are electron deficient or positively charged species.
85. (b)
86. (b) Methyl carbanion is sp^3 hybridised, with three bond pairs and one lone pair same is the case with NH_3 .
87. (d) $-NO_2$ group is electron attractive group, so it is able to deactivate the benzene ring.



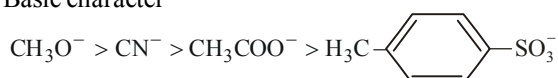
hence withdrawal of electrons from ortho and para position cause easy removal of $-Cl$ atom due to development of +ve charge on o - and p positions.

88. (d) Optical inactivity due to internal compensation is exhibited by meso-tartaric acid. Option (d) is correct.
89. (a) Electron withdrawing group ($-NO_2$) increases acidity while electron releasing group ($-CH_3$, $-H$) decreases acidity. Also effect will be more if functional group present at para position then ortho and then meta.
90. (d) Absolute alcohol cannot be obtained by simple fractional distillation because a Constant boiling azeotropic mixture is formed. It is impossible to get pure ethanol by distilling any mixture of ethanol and water containing less than 95.6% of ethanol. This particular mixture of ethanol and water boils as if it were a pure liquid. It has a constant boiling point, and the vapour composition is exactly the same as the liquid.
91. (c) Simple distillation is suitable for liquids which boil without decomposition at atmospheric pressure. In case of liquids which decompose before their boiling point is reached, vacuum distillation is used. Glycerine decomposes at its boiling point ($298^\circ C$) but can be distilled unchanged at 12 mm pressure at $180^\circ C$.
92. (d) In aromatic acids presence of electron withdrawing substituent e.g. $-NO_2$ disperses the negative charge of the anion and stabilises it and hence increases the acidity of the parent benzoic acid. Further o -isomer will have higher acidity than corresponding m and p isomers. Since nitro group at

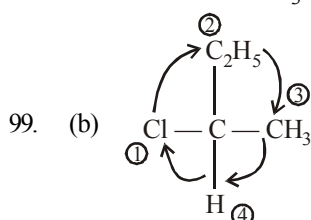
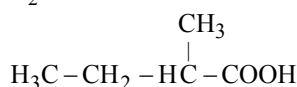
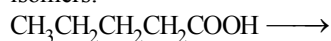
p-position have more pronounced electron withdrawing than $-\text{NO}_2$ group at m-position hence the correct order is the one given above.



93. (b) The stronger the base the more is the nucleophilic character and vice versa.
Basic character

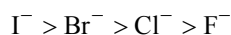


94. (a) 1,2-dihydroxybutane has longest carbon atom chain of 4 atoms and OH groups attached on it at first and second carbon atoms. So the choice (a) is correct.
95. (b) $-\text{I}$ group destabilises carbocation and since inductive effect decreases with increasing length of carbon chain. Therefore (b) is the correct option.
96. (b) Being immiscible liquids, they form different layers hence can be separated with the help of separating funnel.
97. (c) Simple distillation can be applied for the separation and purification of a mixture of two or more liquids provided their boiling points differ at least by 30 K. In such cases, the low boiling (more volatile) liquid of the mixture will distil over first. The temperature of the distillation flask is then raised when high boiling (less volatile) liquid of the mixture will distil, it is collected in a separate flask.
98. (c) Pentanoic acid and 2-methyl butanoic acid are chain isomers.

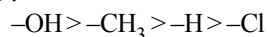


R-configuration

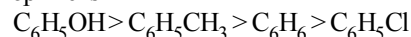
100. (a) Nucleophilicity increases down the periodic table.



101. (c) Electrophiles have high affinity for electrons. They attack at the site where electron-density is highest. Electron donating groups increases the electron density. The electron donating tendency decreases in the order :



Therefore, the correct order of reactivity towards electrophile is

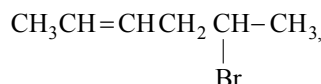


102. (c) Out of the given compounds the most reactive towards

electrophilic attack is Phenol forms phenoxide

ion which is stable due to resonance i.e. the correct answer is option (c).

103. (a) In the molecule



the number of stereoisomers is given by sum of geometrical isomers (because of presence of $\text{C}=\text{C}$) and optical isomers (because of presence of chiral carbon atom).

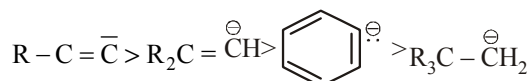
Number of geometrical isomers = 2 (one $\text{C}=\text{C}$ is present).

Number of optical isomers = 2 (one chiral carbon atom).

Total number of stereoisomers = $2 + 2 = 4$

104. (d) The stability of carbanions is affected due to resonance, inductive effect and s-character of orbitals. Greater the number of groups having +I group (alkyl group) lesser stable would be the carbanion.

Further stability of carbanion decreases with decrease in s-character. Benzene carbanions are stabilized due to resonance, hence the correct order is



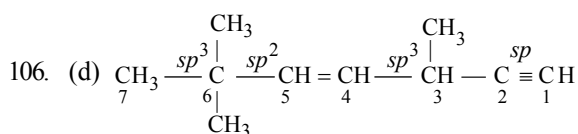
The correct order of stability of given carbanion is in the order I > III > II > IV.

105. (b) C-1 is sp hybridized ($\text{C}\equiv\text{C}$)

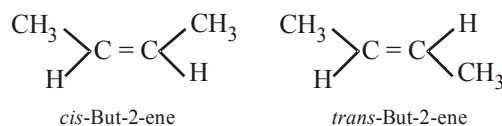
C-3 is sp^3 hybridized ($\text{C}-\text{C}$)

C-5 is sp^2 hybridized ($\text{C}=\text{C}$)

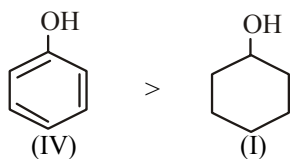
Thus the correct sequence is sp, sp^3, sp^2 .



107. (d) Alkenes with double bonds cannot undergo free rotation and can have different geometrical shapes with two different groups on each end of the double bond.



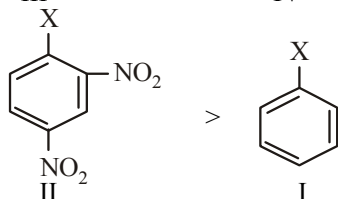
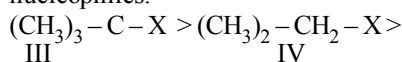
108. (a)
-
- (III) (II)



Explanation : Presence of three —NO₂ groups in *o*-, *p*- positions to phenolic groups (in III) makes phenol strongly acidic because its corresponding conjugate base is highly stabilised due to resonance. Conjugate base of CH₃COOH, II (*i.e.* CH₃COO⁻) is resonance hybrid of two equivalent structures. The conjugate base of phenol, IV is stabilized due to resonance (note that here all resonating structures are not equivalent). The conjugate base of cyclohexanol, I does not exhibit resonance, hence not formed.

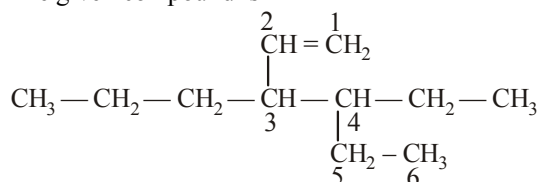
109. (a) Tertiary alkyl halide is most reactive towards nucleophilic substitution because the corresponding carbocation (3°) is most stable. Aryl halide is least reactive due to partial double bond character of the C — Cl bond.

Presence of — NO₂ groups in *ortho* and *para* positions increases the reactivity of the — Cl towards nucleophiles.



or I < II < IV < III

110. (d) Cl⁻ is the weakest base and hence better leaving group
111. (a) The given compound is

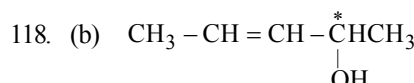
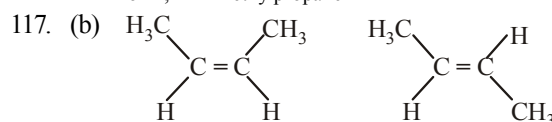
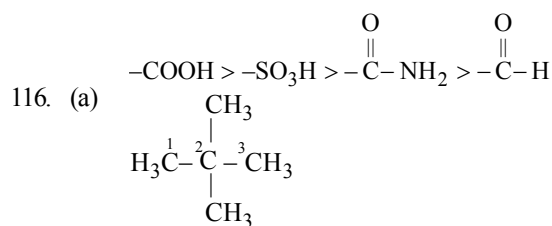


112. (a)

Correct IUPAC name of above compound is trans-2-chloro-3-iodo-2-pentene

113. (a)
- 3-ethyl-4,4-dimethyl heptane

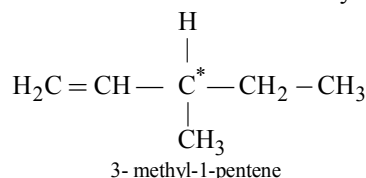
114. (a) Nitro group is electron withdrawing group, so it deactivates the ring towards electrophilic substitution.
115. (a) The correct order of priority for the given functional group is



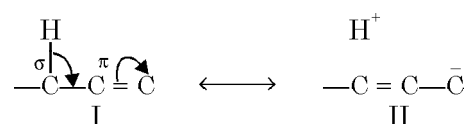
exhibits both geometrical as well as optical isomerism.

cis - R *cis* - S
trans - R *trans* - S

119. (c) For a compound to show optical isomerism, presence of chiral carbon atom is a necessary condition.

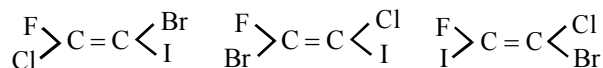


120. (b) —CN has highest priority. Further the sum of locants is 7 in (b) and 9 in (d).
121. (b) Alkyl groups with at least one hydrogen atom on the α-carbon atom, attached to an unsaturated carbon atom, are able to release electrons in the following way.



Note that the delocalisation involves σ and π bond orbitals (or *p* orbitals in case of free radicals) ; thus it is also known as σ — π conjugation. This type of *electron release due to the presence of the system* H—C—C=C is known as hyperconjugation

122. (b) Allene (C₃H₄) is $\text{H}_2\overset{sp^2}{\text{C}}=\overset{sp}{\text{C}}=\overset{sp^2}{\text{CH}_2}$
123. (b) Chiral conformation will not have plane of symmetry. Since twist boat does not have plane of symmetry it is chiral.
124. (d) Methyl group can't produce electromeric and resonance effects. Inductive effect affects all the three positions of the ring in decreasing order of *ortho* > *meta* > *para*. It shows hyperconjugation (no-bond resonance) which explains beautifully the *o*, *p*-directing influence of the —CH₃ group.
125. (d) Taking any two halogens, the possible structural isomers for the alkene can be three.



Each of the three alkene can exist as *E* and *Z* isomer, making total number of isomers as six.